Protein Structure

- Tools and databases
- Methods

What we hope to learn

- The basic building blocks: amino acids
- Secondary structure
- Forces that drive folding
- Motifs or supersecondary structure
- Domains
- Finding out more about structures

Why we should care *more* about structure and *less* about informatics…

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Structure *and* dynamics are needed to infer function.

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There is more to bioinformatics than AGCT or protein sequence

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There is more to bioinformatics than AGCT or protein sequence

RNA is more important than DNA

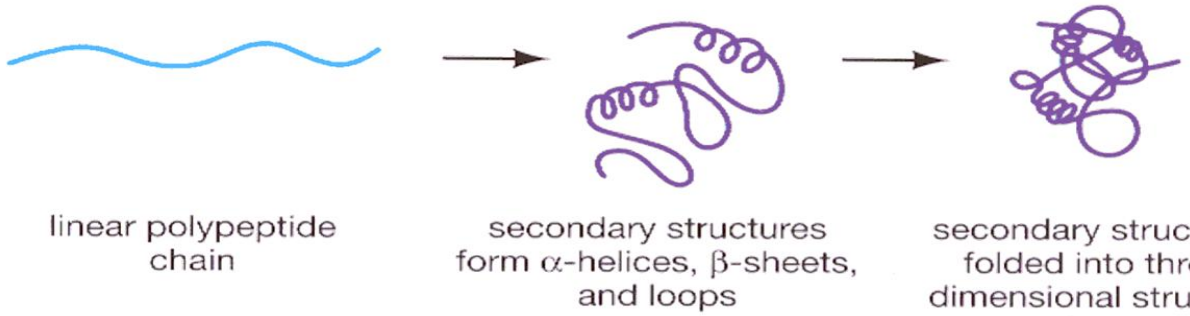
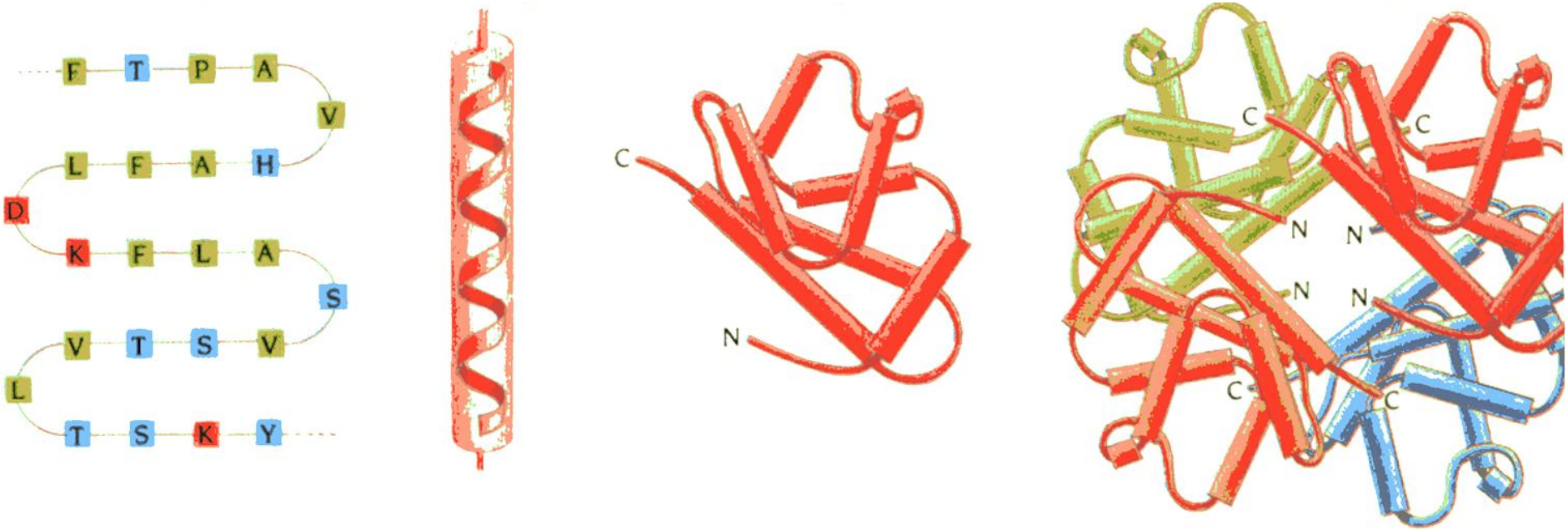


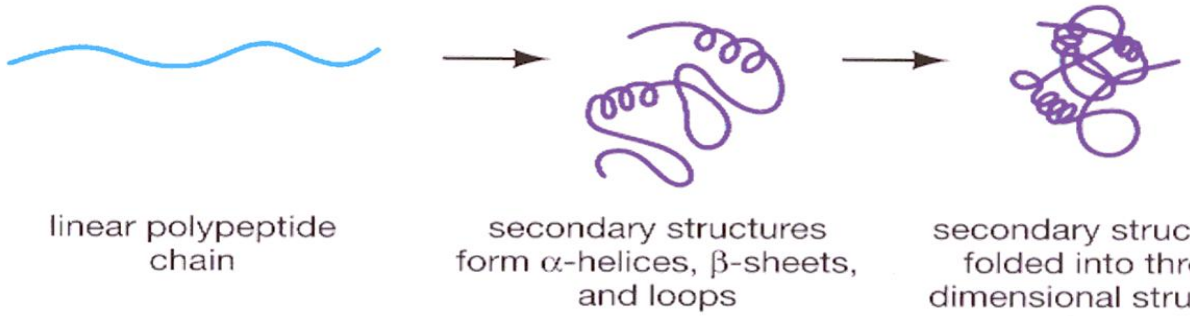
Figure 9.1. Pathway for folding a linear chain of amino acids into a three-dimensional protein structure.

(from Mount, "Bioinformatics Sequence and Genome Analysis"

primary (1º) secondary (2º) tertiary (3º) quaternary (4º)



From: Brandon & Tooze, "Introduction to Protein Structure"



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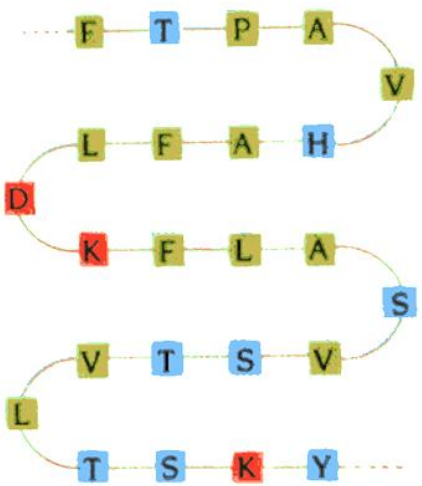

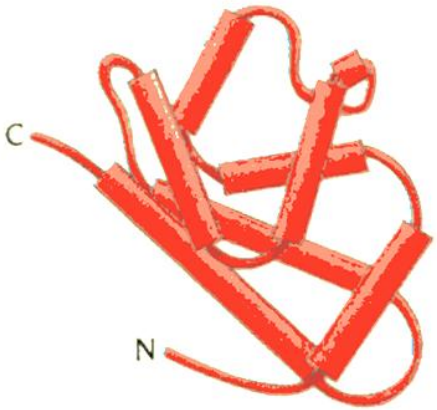

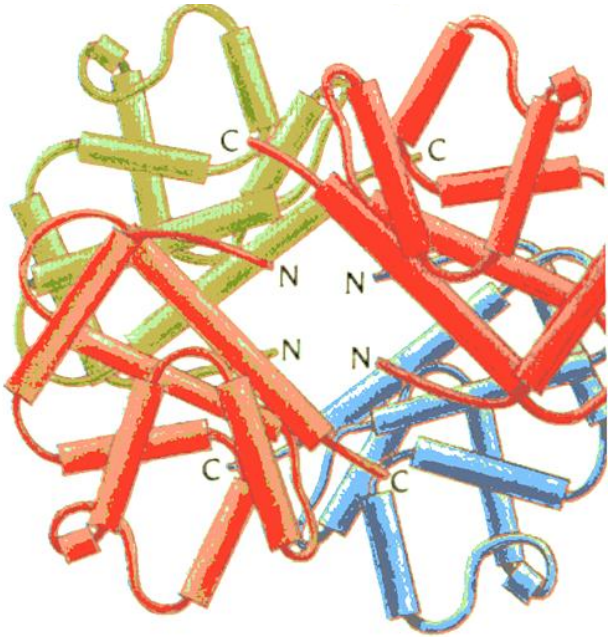

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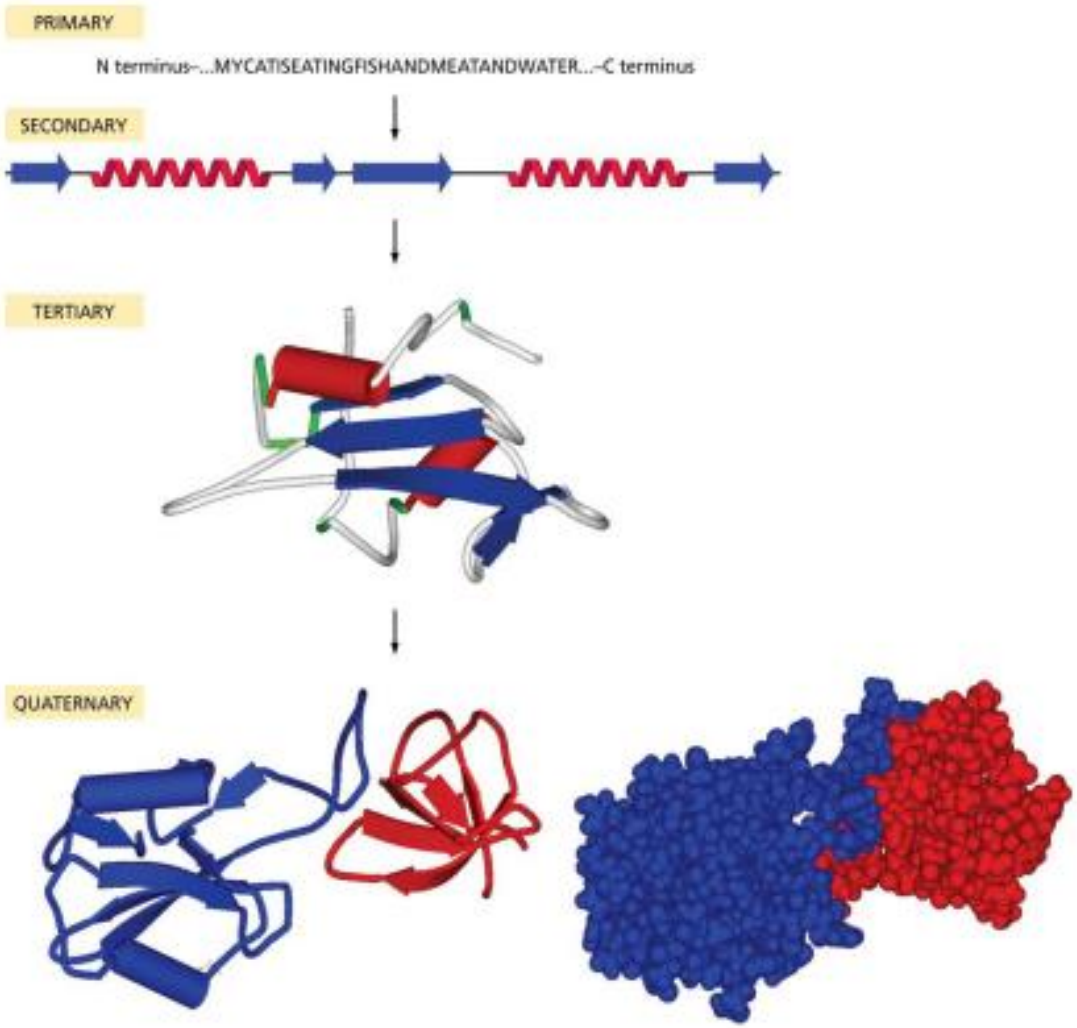
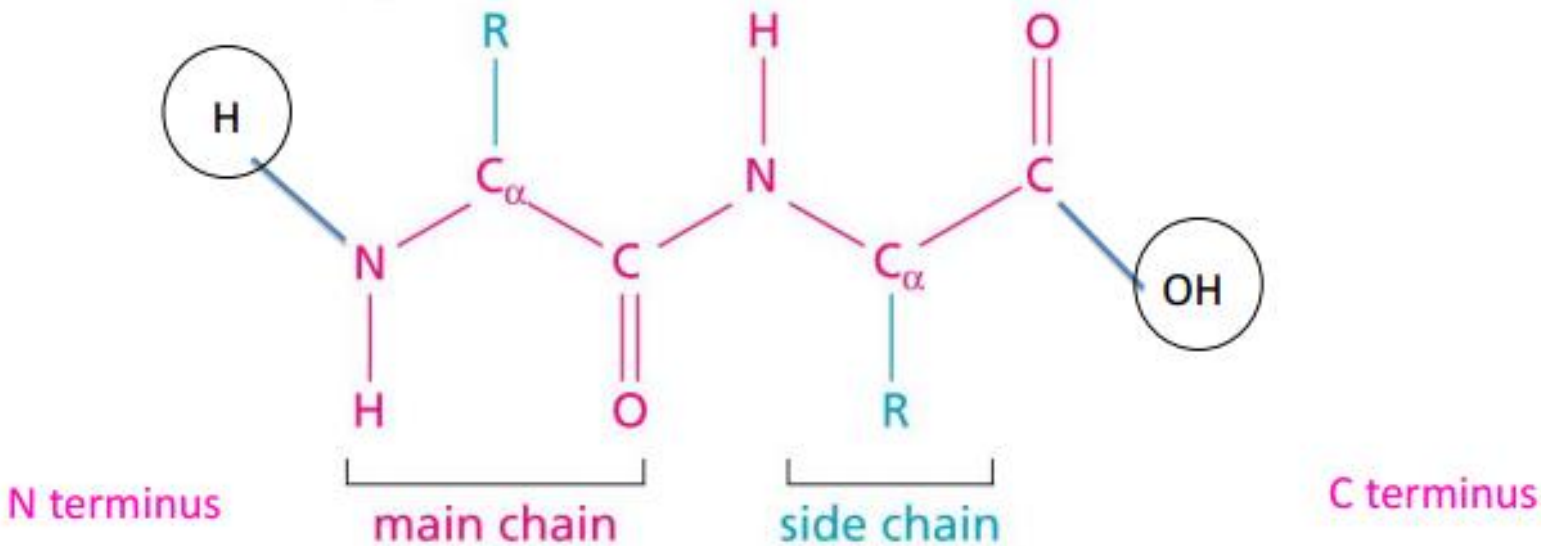
Proteins are linear polymers that fold up by themselves…mostly



The parts of a protein



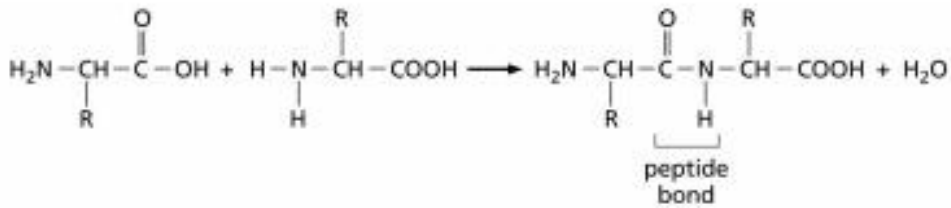
N, C, C, N, C, C… : "Backbone"

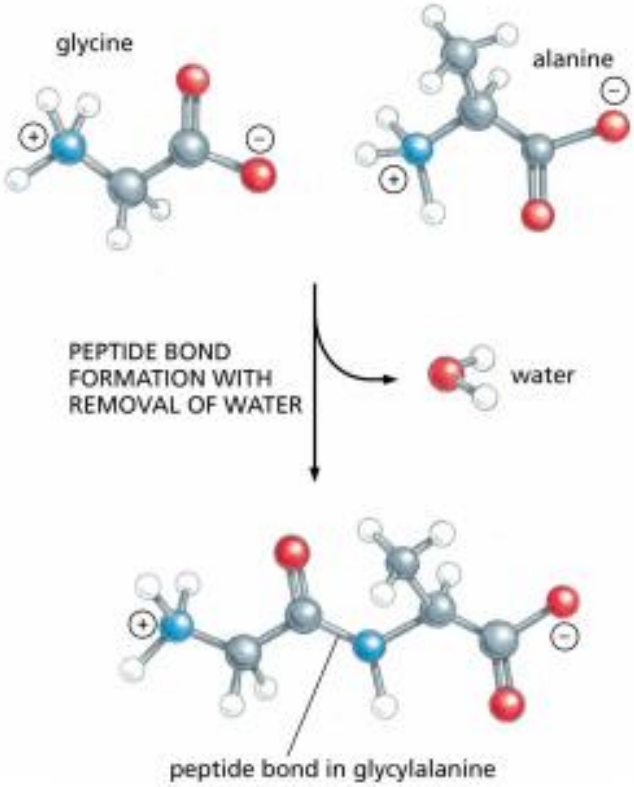
R: "side chain"

C, N, H, O, S

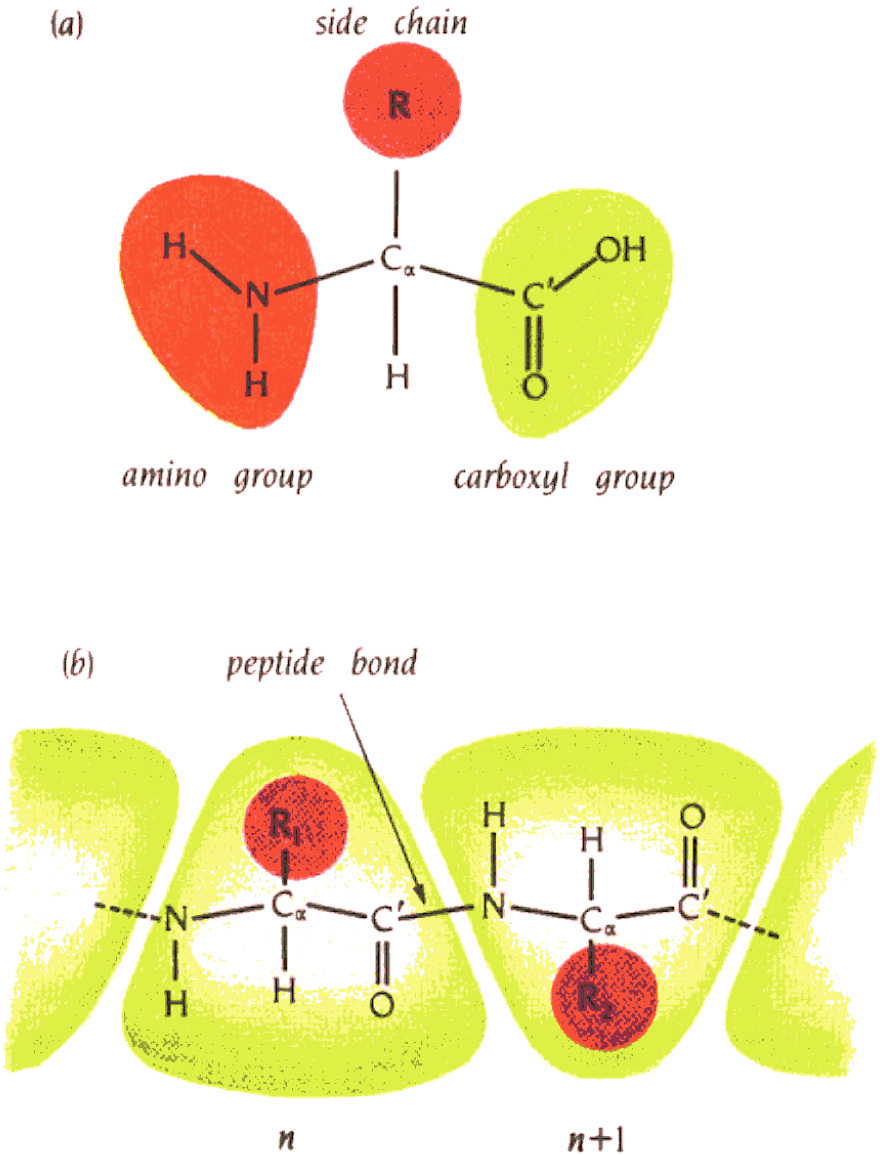
polymers of amino acids

Polypeptide





- Bond between carbonyl carbon to amine nitrogen – **peptide bond**
- Two or more amino acids attached by a peptide bond - **polypeptide**
- In aqueous solution, C-terminus loses a proton, N-terminus picks up a proton



amino acids/proteins:

Glycine	GLY	G
Alanine	ALA	A
Phenylalanine	PHE	F
Leucine	LEU	L
Isoleucine	ILE	I
Valine	VAL	V
Proline	PRO	P
Methionine	MET	M
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Threonine	THR	T
Tyrosine	TYR	Y
Tryptophan	TRP	W
Histidine	HIS	H
Cysteine	CYS	C

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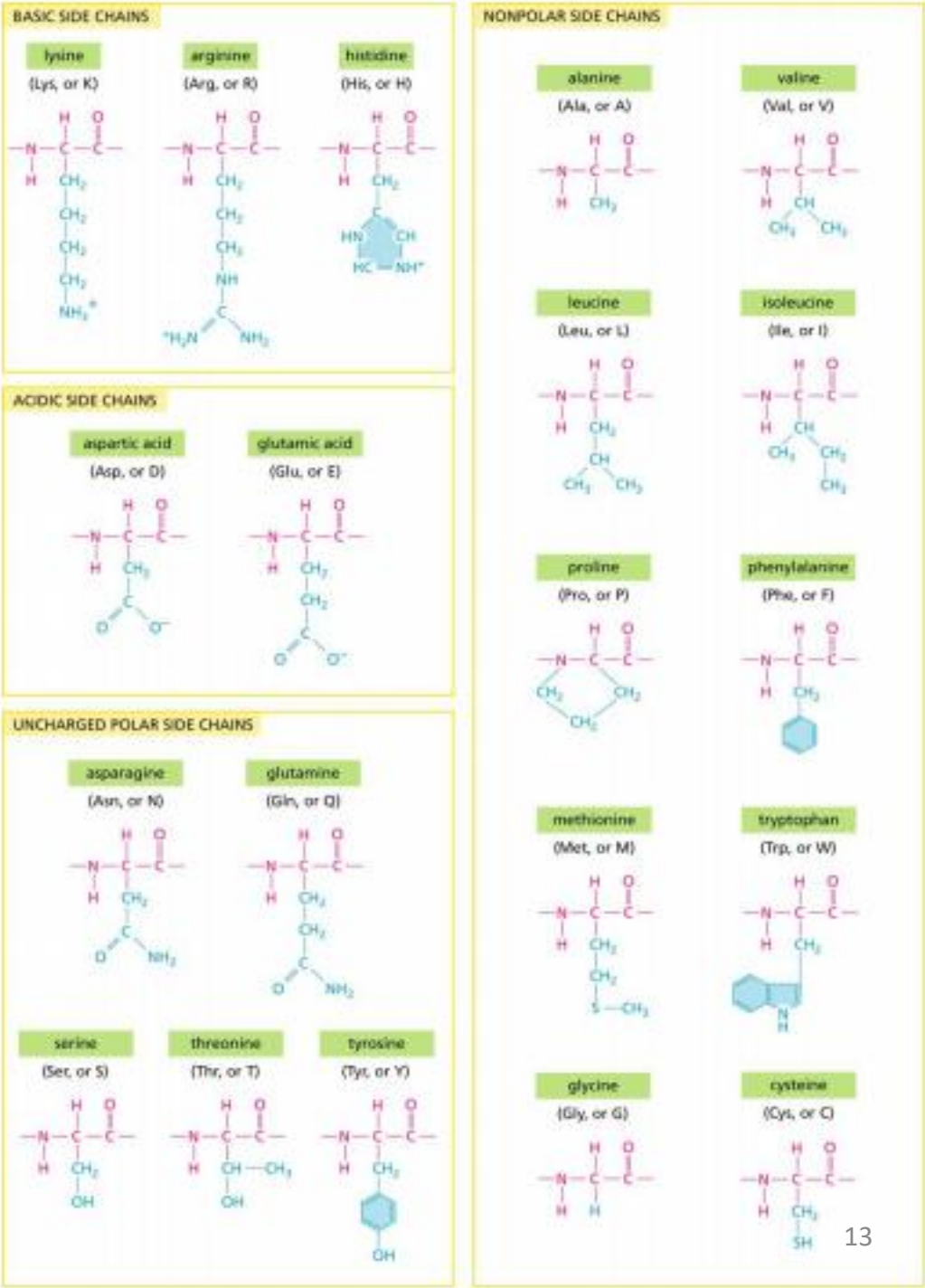
The amino acids

- Can be grouped by properties in many ways according to the chemical and physical properties (e.g. size) of the side chain.

- One grouping based on chemical properties:
  - Basic: proton acceptors
  - Acidic: proton donors
  - Uncharged polar: have polar groups like $CONH_2$ or CH_2OH
  - Nonpolar: tend to be hydrophobic
  - Weird: proline links to the N in the main chain
  - Strong: Cysteine can make "disulphide bridges"



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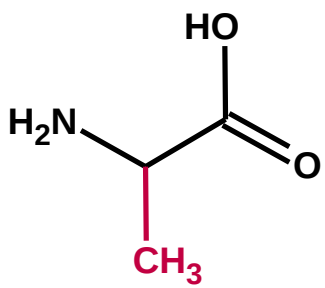
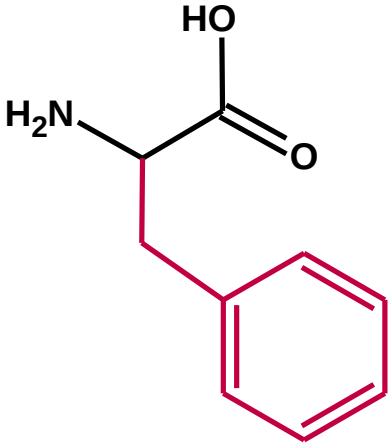
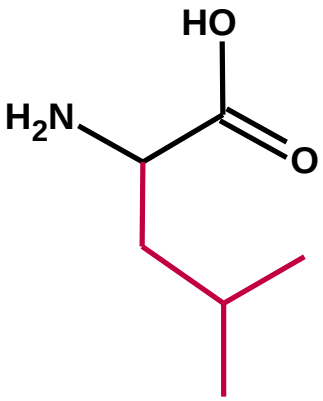


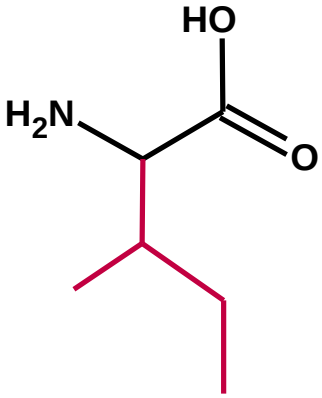
Alanine



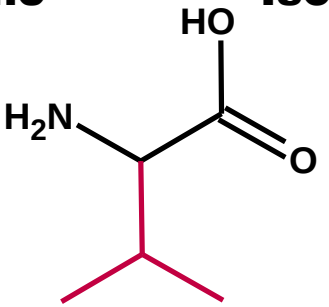
Phenylalanine



Leucine



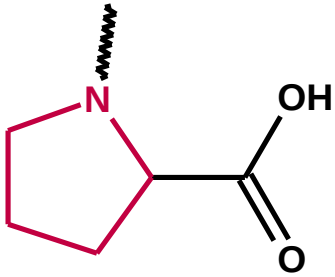
Isoleucine



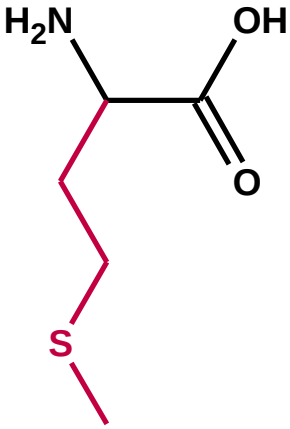
Valine

Hydrophobic amino acids

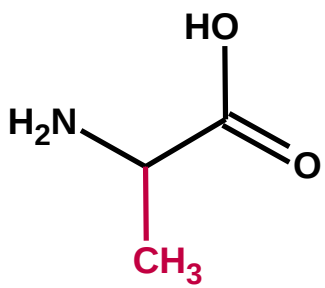
Alanine	ALA	A
Phenylalanine	PHE	F
Leucine	LEU	L
Isoleucine	ILE	I
Valine	VAL	V
Proline	PRO	P
Methionine	MET	M



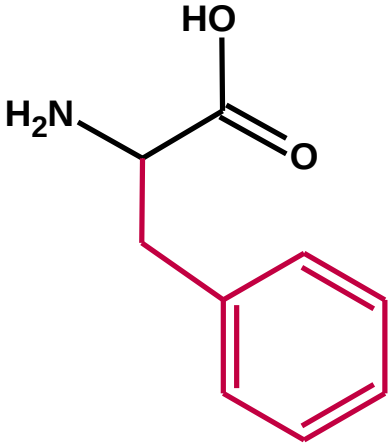
Proline



Methionine



Alanine



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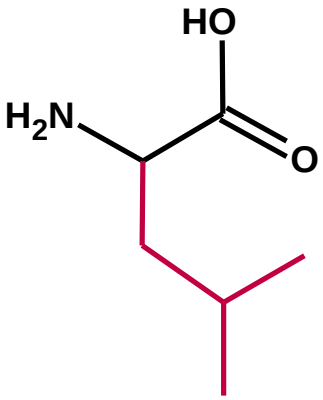
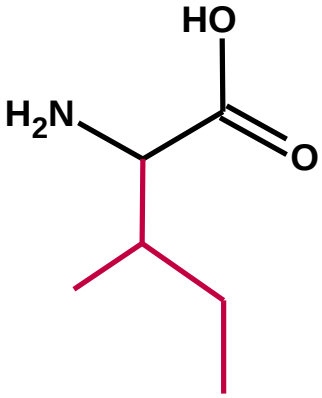


Leucine



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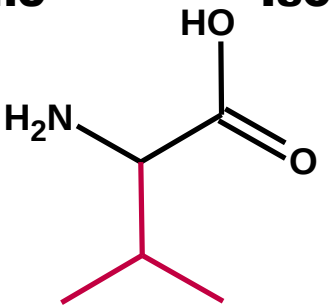
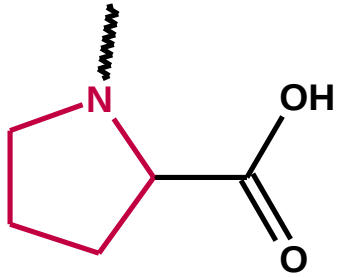
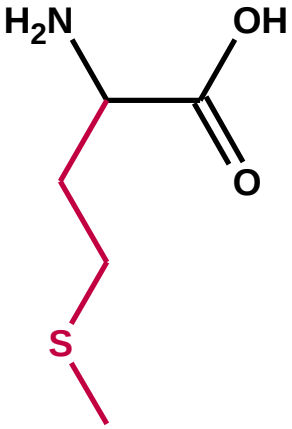


Valine



Proline

Helix breaker
(except at N-caps)



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Tyrosine	TYR	Y
Tryptophan	TRP	W
Histidine	HIS	H
Cysteine	CYS	C

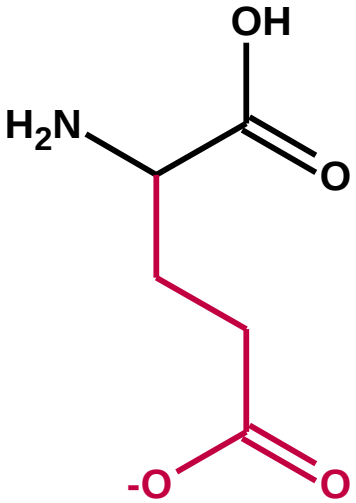
Charged amino acids

?

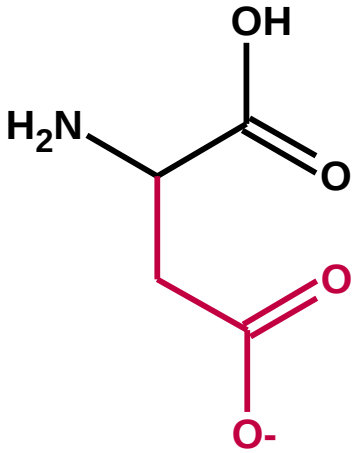
Charged residues tend to reside on protein surface

Charged amino acids:

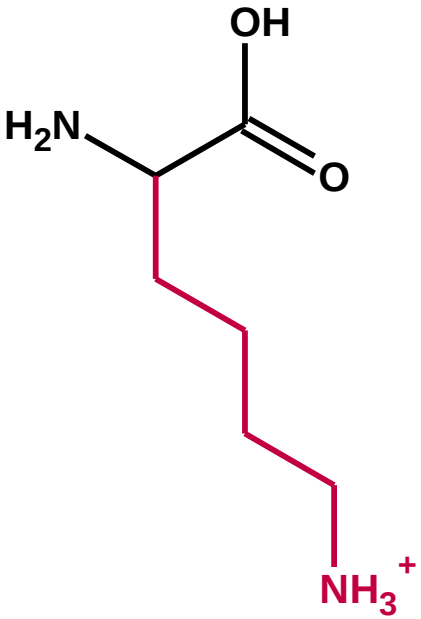
Glutamic acid	GLU	E
Aspartic acid	ASP	D
Lysine	LYS	K
Arginine	ARG	R
Histidine	HIS	H



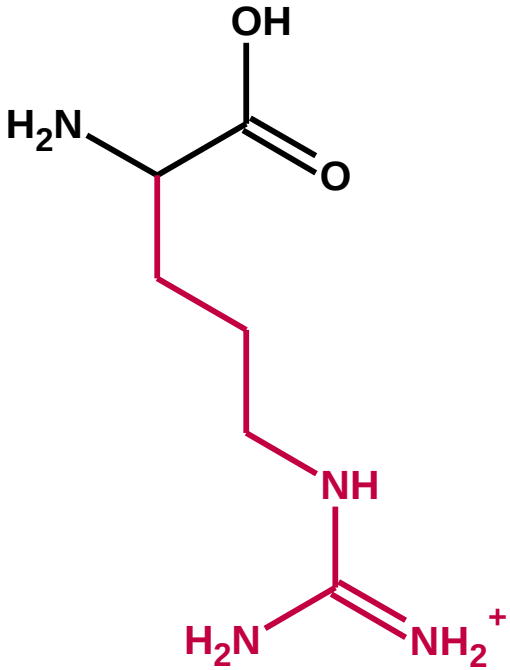
Glutamic acid



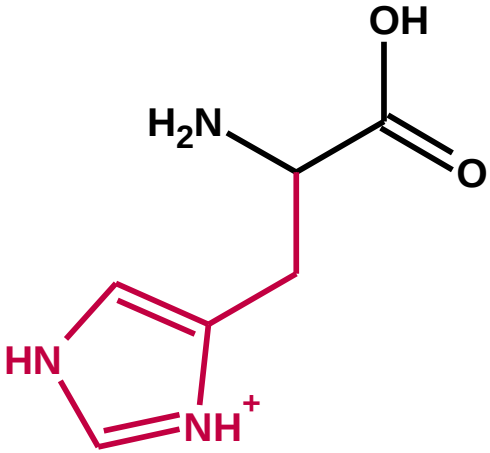
Aspartic acid



Lysine



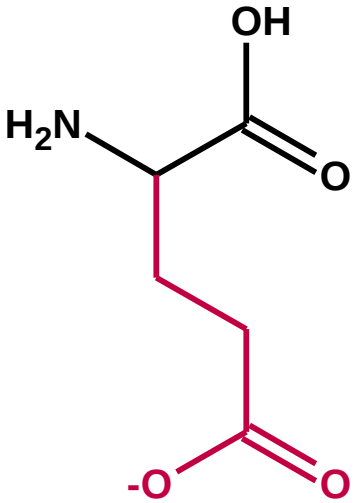
Arginine



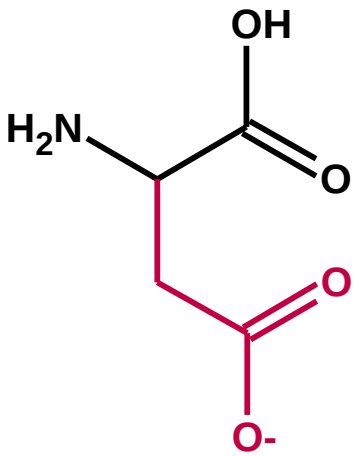
Histidine

Charged amino acids:

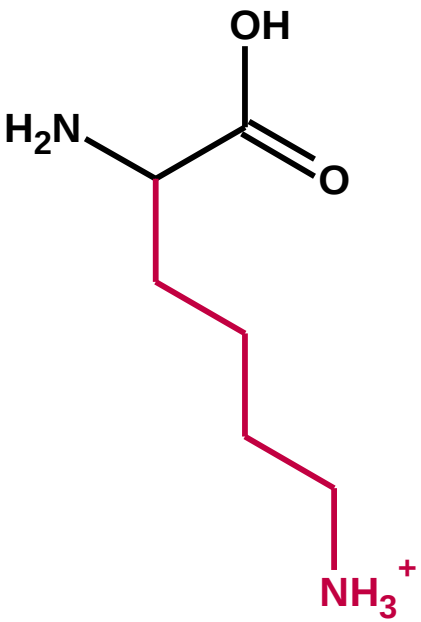
Glutamic acid	GLU	E
Aspartic acid	ASP	D
Lysine	LYS	K
Arginine	ARG	R
Histidine	HIS	H



Glutamic acid



Aspartic acid



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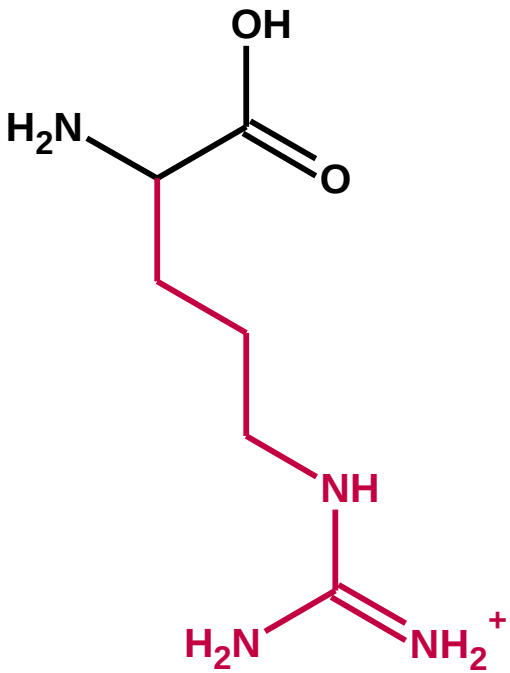
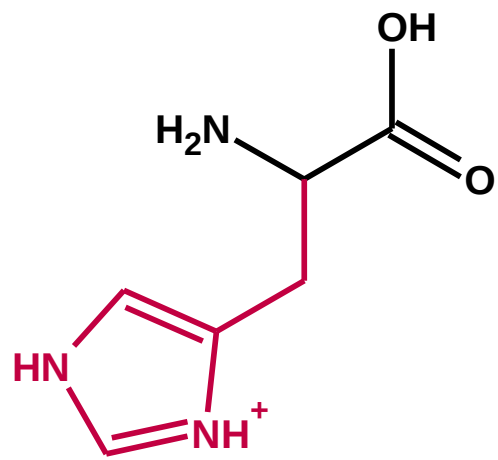


Arginine



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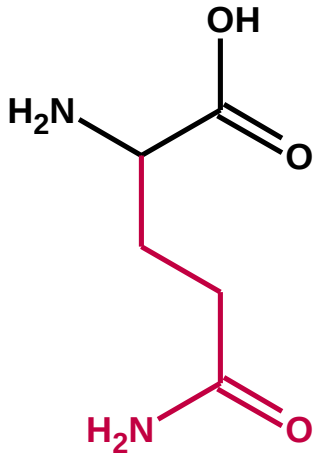
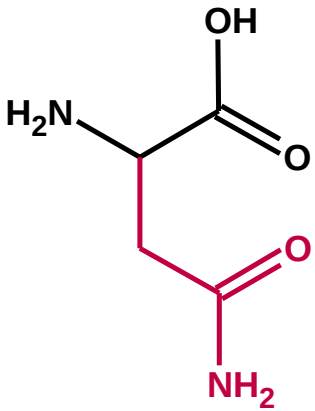
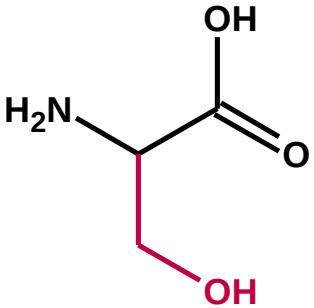


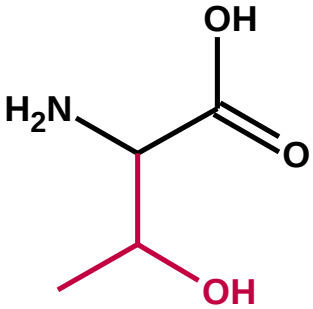
Glutamine



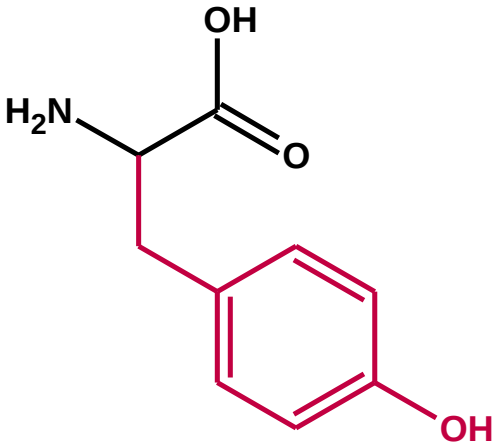
Asparagine



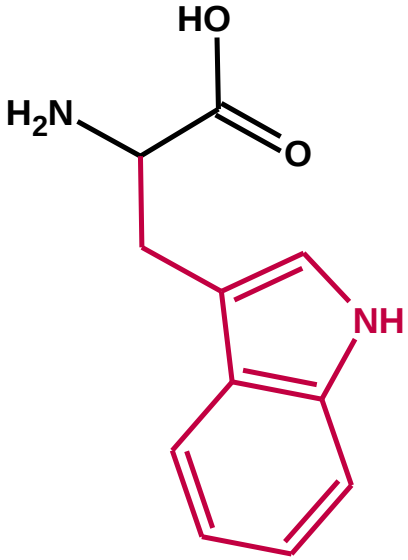
Serine



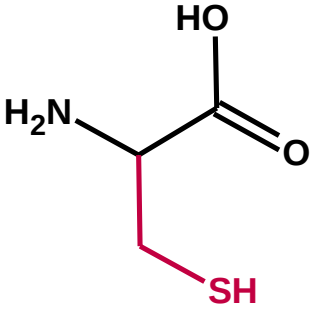
Threonine



Tyrosine



Tryptophan



Cysteine

Classification of amino acids

- **Amino acids with nonpolar side chains**: They -carry non-ionized, nonpolar, hydrophobic (water-repellant), hydrocarbon side chains. They are mostly *mono amino mono carboxylic* acids having a single NH_2 and COOH group on the α-carbon. The non polar side chain of two such amino acids may form hydrophobic bonds with one another. Its further classified as
  - Amino acids with non polar aliphatic side chains: (A,L,I,M)
  - Amino acids with non polar aromatic side chains: (F,W)
  - Imino acids: (P)
- **Amino acids with polar side chains:** They carry side chains bearing ionized or non-ionized hydrophilic (water-attracting) polar groups and classified as
  - Amino acids with uncharged polar side chain: OH groups,Sulfur groups, amides groups
  - Amino acids with charged polar side chain: groups like guanidinium or imidazole group.

Continued..

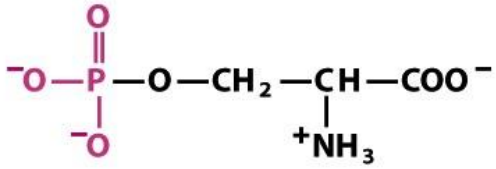
- Amino acids with +vely charged side chains bear a cationic group NH_3^+ in the side chain. It includes histidine with an imidazole group, arginine with guanidine group and lysine with diamino mono carboxylic acid having an NH_3^+

- Amino acids with –vely charged side chains bear an anionic group COO^- in the side chain, which include glutamic and aspartic acids which are mono amino dicarboxylic acids.

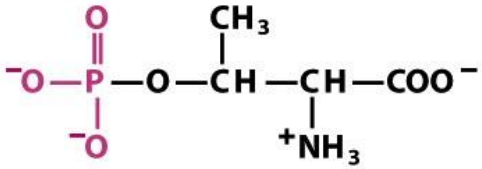
Amino Acids not occurring in proteins

- They occur either in free forms or in specific complexes.
- They include β-alanine (coenzyme A, Vitamins)
- Dopa as neurotransmitter (melanin precursor), and 3-hydroxy proline

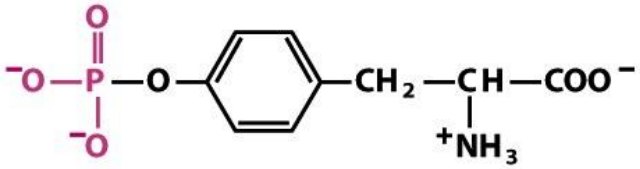
Uncommon Amino Acids in Proteins



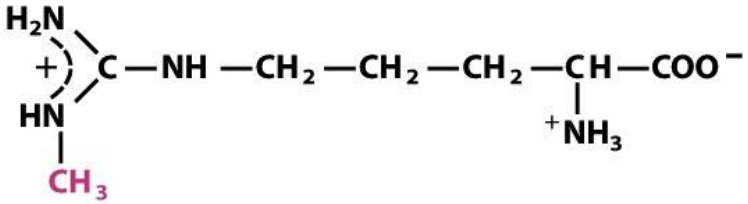
Phosphoserine



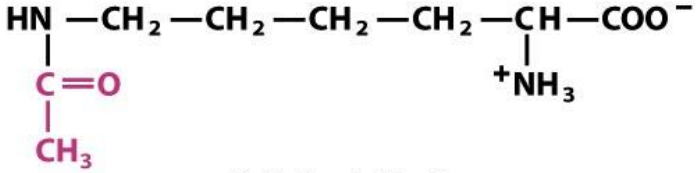
Phosphothreonine



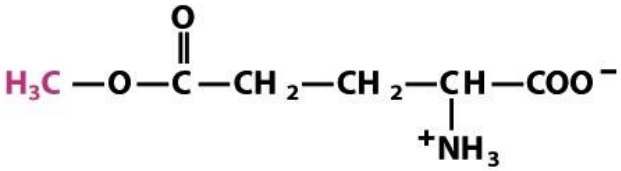
Phosphotyrosine



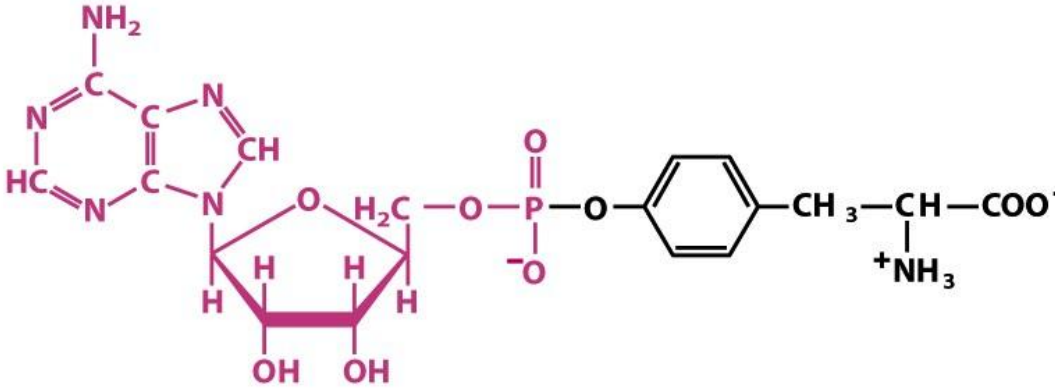
σ-N-Methylarginine



6-*N*-Acetyllysine



Glutamate methyl ester



Adenylyltyrosine

Figure 3-8b

Lehninger Principles of Biochemistry, Fifth Edition

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Properties of Amino acids

- Amino acids are colorless, crystalline, water-soluble solids with high melting points.
- Depending upon the pH, the α-NH_2 group and side chain groups such as NH_2, imidazole and guanido groups may accept α- H^+ to form cations while the α COOH and side chain groups may donate H^+ to form anions.
- They consequently behaves as an ampholyte (acting as donor and accepter)
- At a specific pH it is called isoelectric pH (pI), it called zwitterion carrying equal number of positive and negative charges and net charge will be zero

Amino Acids Carry a Net Charge of Zero at a Specific pH

- Zwitterions predominate at pH values between the pK_a values of amino and carboxyl group
- For amino acid without ionizable side chains, the Isoelectric Point (equivalence point, pI) is

$$pI = \frac{pK_1 + pK_2}{2}$$

- At this point, the net charge is zero
  - AA is least soluble in water
  - AA does not migrate in electric field

Ionization

At acidic pH, the carboxyl group is protonated and the amino acid is in the cationic form

At neutral pH, the carboxyl group is deprotonated but the amino group is protonated. The net charge is zero; such ions are called Zwitterions

At alkaline pH, the amino group is neutral $-NH_2$ and the amino acid is in the anionic form.

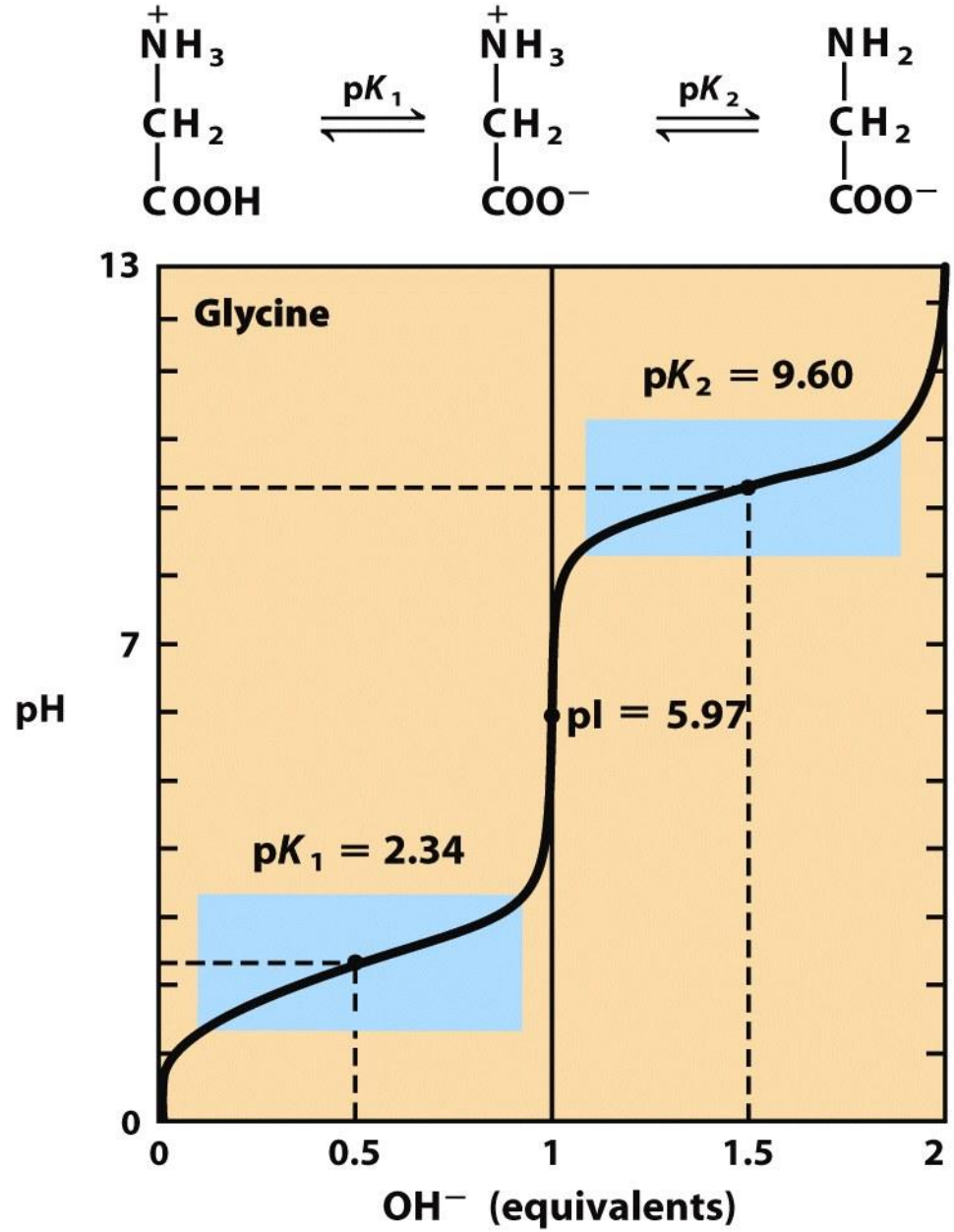

Figure 3-10
Lehninger Principles of Biochemistry, Fifth Edition
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Substituent effects on pK_a Values

α-carboxy group is much more acidic than in carboxylic acids
α-amino group is slightly less basic than in amines

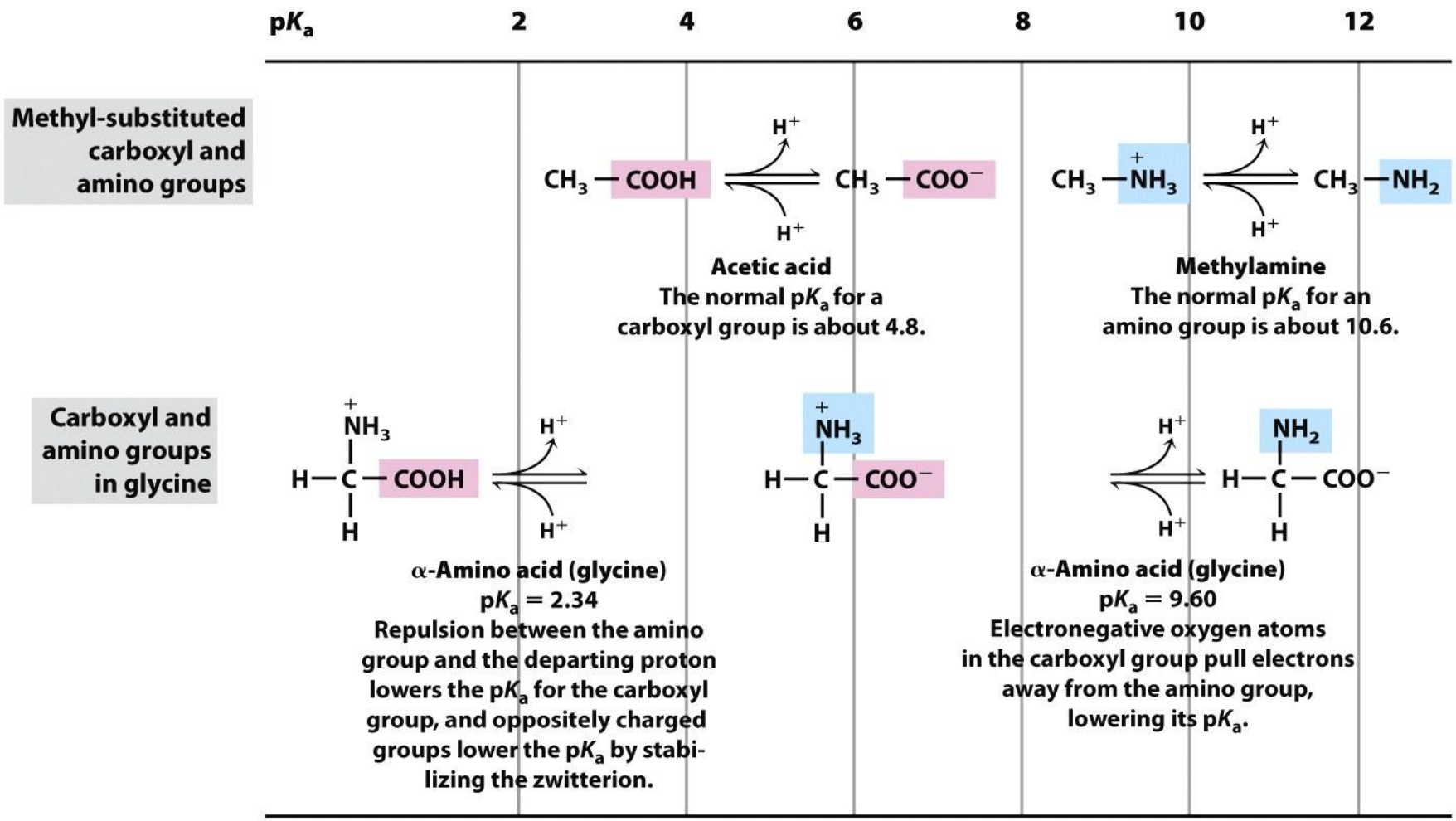

Figure 3-11
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Amino Acids Can Act as Buffers

Amino acids with uncharged side-chains, such as glycine, have two pK_a values:

The pK_a of the α-carboxyl group is 2.34

The pK_a of the α-amino group is 9.6

It can act as a buffer in two pH regimes.

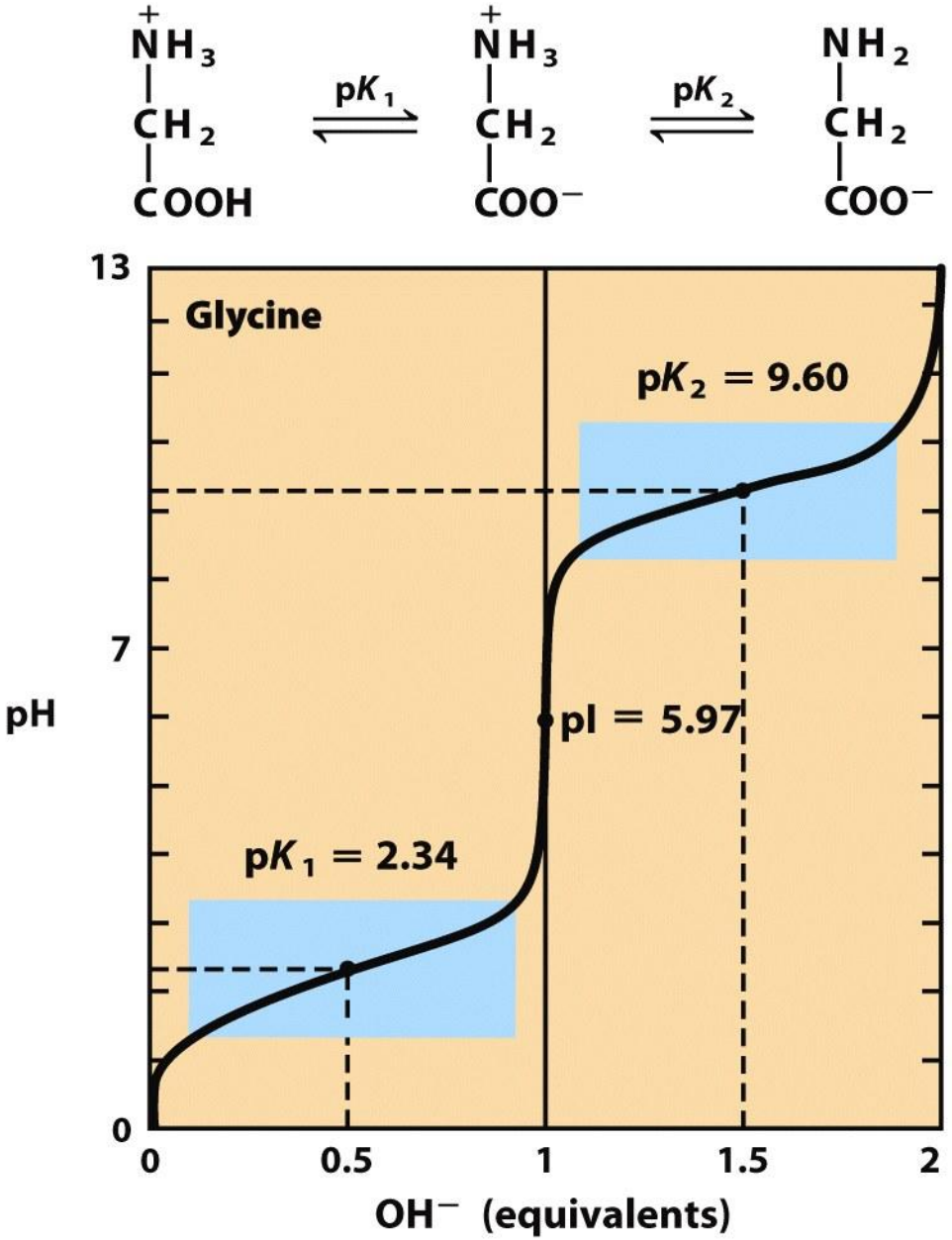

Figure 3-10
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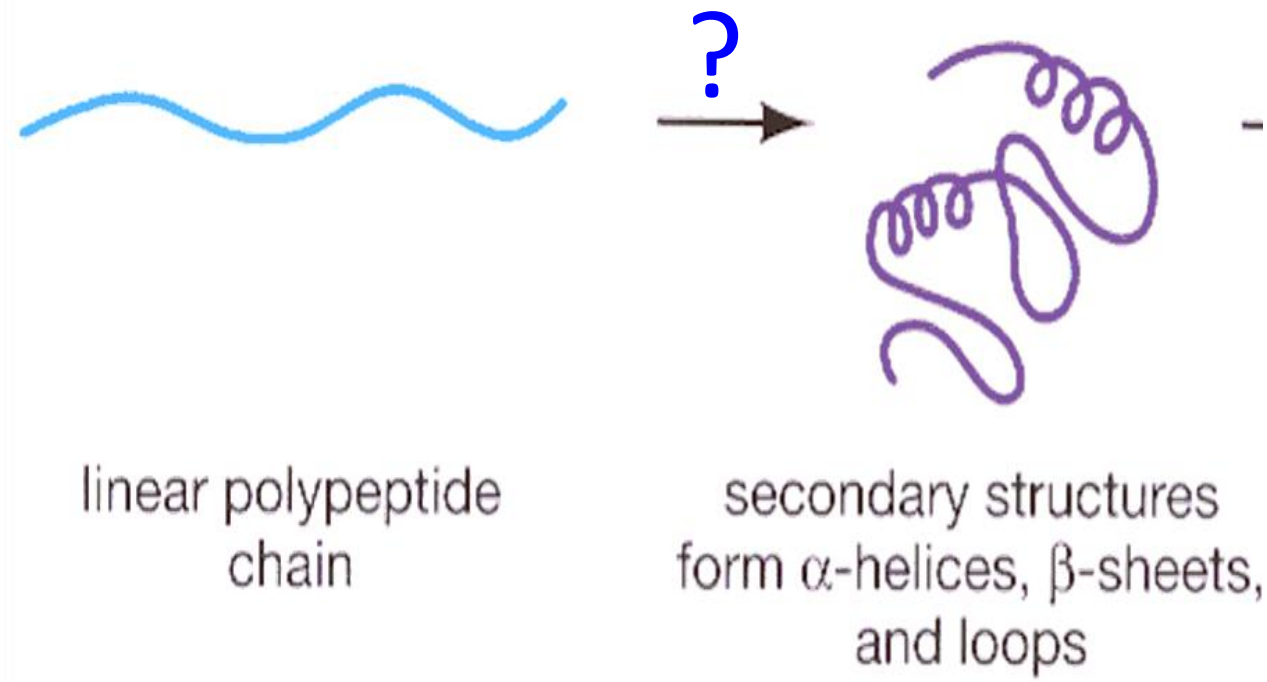
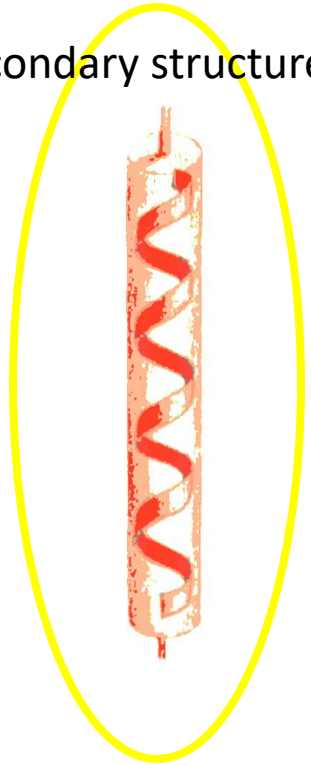

?
linear polypeptide chain
secondary structures form α-helices, β-sheets, and loops

amino acids/proteins:

ala, gly, leu, ile, val, glu, asp, gln, asn, pro,
lys, arg, ser, thr, tyr, trp, phe, his, cys, met

Secondary structure



α-helix
β-sheet
loop or turn
random coil

From: Brandon & Tooze, “Introduction to Protein Structure”

Can we predict 2° structure?

Can we predict 2° structure?

-- helical propensities
-- mining structural databases for known sequence-structure relationships
-- sequence contexts?
-- neural nets, …

Yes!

…but accuracy is poor ☹

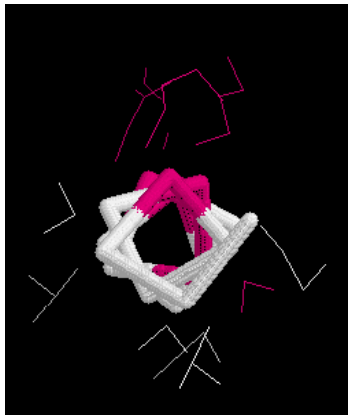
However: a given sequence may adopt multiple structures.

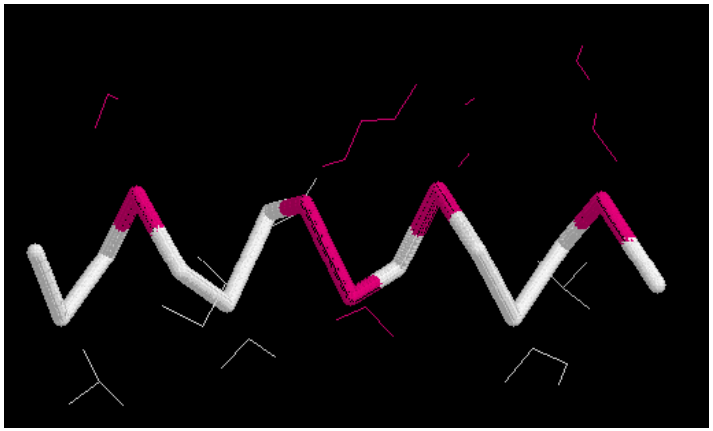
Kabsch W & Sander C *Proc. Natl. Acad. Sci.* 81, 1075-1078 (1984)
"On the use of sequence homologies to predict protein structure: Identical pentapeptides can have completely different conformations"

Can we predict properties of unknown Proteins?

Can we predict properties of unknown Proteins?

Yes!





Sequence: LAKMVVKTAEAILKD

α Helix

3.6 amino acids per turn

10 Å (3 turns) 10 amino acids

N-H•••O=C H-bond between every 4th residue

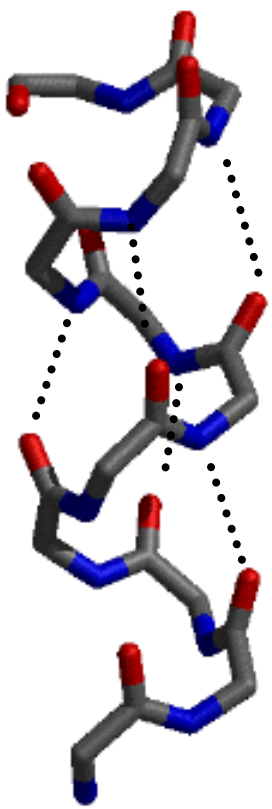
Transmembrane α Helix ≈ 19 hydrophobic AA long

Parametric Sequence Analysis

http://www.expasy.org/tools/

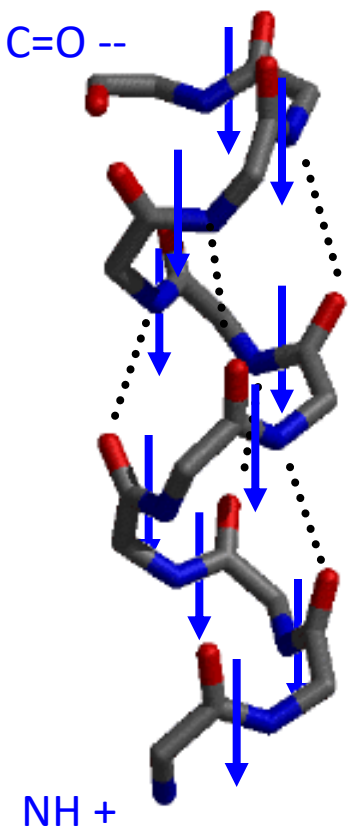
- Parameters assigned to residues
  - Amino acids, dipeptides etc.
  - Bases, dinucleotides, triplets etc.
- Values averaged over a sliding window (or pattern)
  - Arithmetic average, geometric average, etc.
  - Values can be weighted based on position in the window
- Threshold learned from training sequences
  - Requires a gold standard known to belong to set
  - Requires sequences known not to belong to set

Example: Kyte-Doolittle assigned a value to each amino acid based on its hydrophobicity



α-helix

i to *i*+4 connectivity
3.6 residues per turn



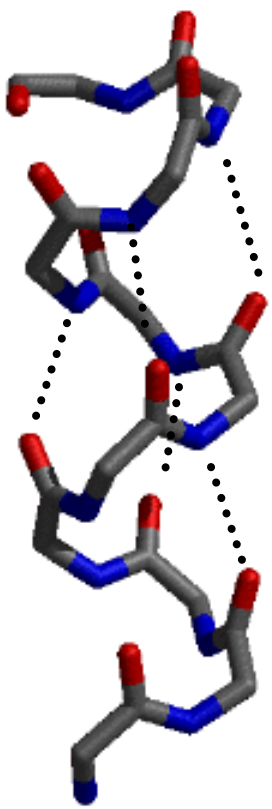
α-helix

i to *i*+4 connectivity
3.6 residues per turn

All NH and C=O groups are connected
except for first NH group and last C=O

⇒ ends are polar (hence usually at protein surface)

⇒ orientation of h-bonds leads to net dipole



α-helix

i to *i*+4 connectivity
3.6 residues per turn

All NH and C=O groups are connected
except for first NH group and last C=O

⇒ ends are polar (hence usually at protein surface)

⇒ orientation of h-bonds leads to net dipole

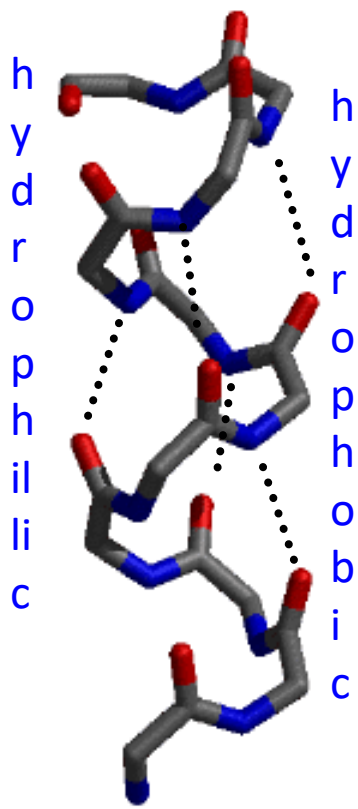
Different side chains have different helical propensities

ALA, GLU, LEU, MET: good α-helix formers

PRO, GLY, TYR, SER: poor α-helix formers

PRO is OK as a N-cap

ASP is a good N-cap (balances +)



α-helix

i to *i*+4 connectivity
3.6 residues per turn

All NH and C=O groups are connected
except for first NH group and last C=O

⇒ ends are polar (hence usually at protein surface)

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Different side chains have different helical propensities

ALA, GLU, LEU, MET: good α-helix formers

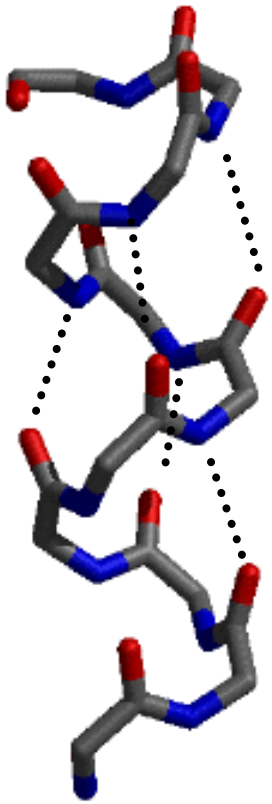
PRO, GLY, TYR, SER: poor α-helix formers

…weak tendency (not strong enough for prediction since hydrophobic residue can be on surface) for amino acids to change from hydrophillic to hydrophobic with a periodicity of 3-4 residues

Left handed helix is also possible,

i to *i*+3: 3_{10} helix

i to *i*+5: π helix



α-helix

i to *i*+4 connectivity

3.6 residues per turn

All NH and C=O groups are connected except for first NH group and last C=O

$\Rightarrow$ ends are polar (hence usually at protein surface)

$\Rightarrow$ orientation of h-bonds leads to net dipole

Different side chains have different helical propensities

ALA, GLU, LEU, MET: good α-helix formers

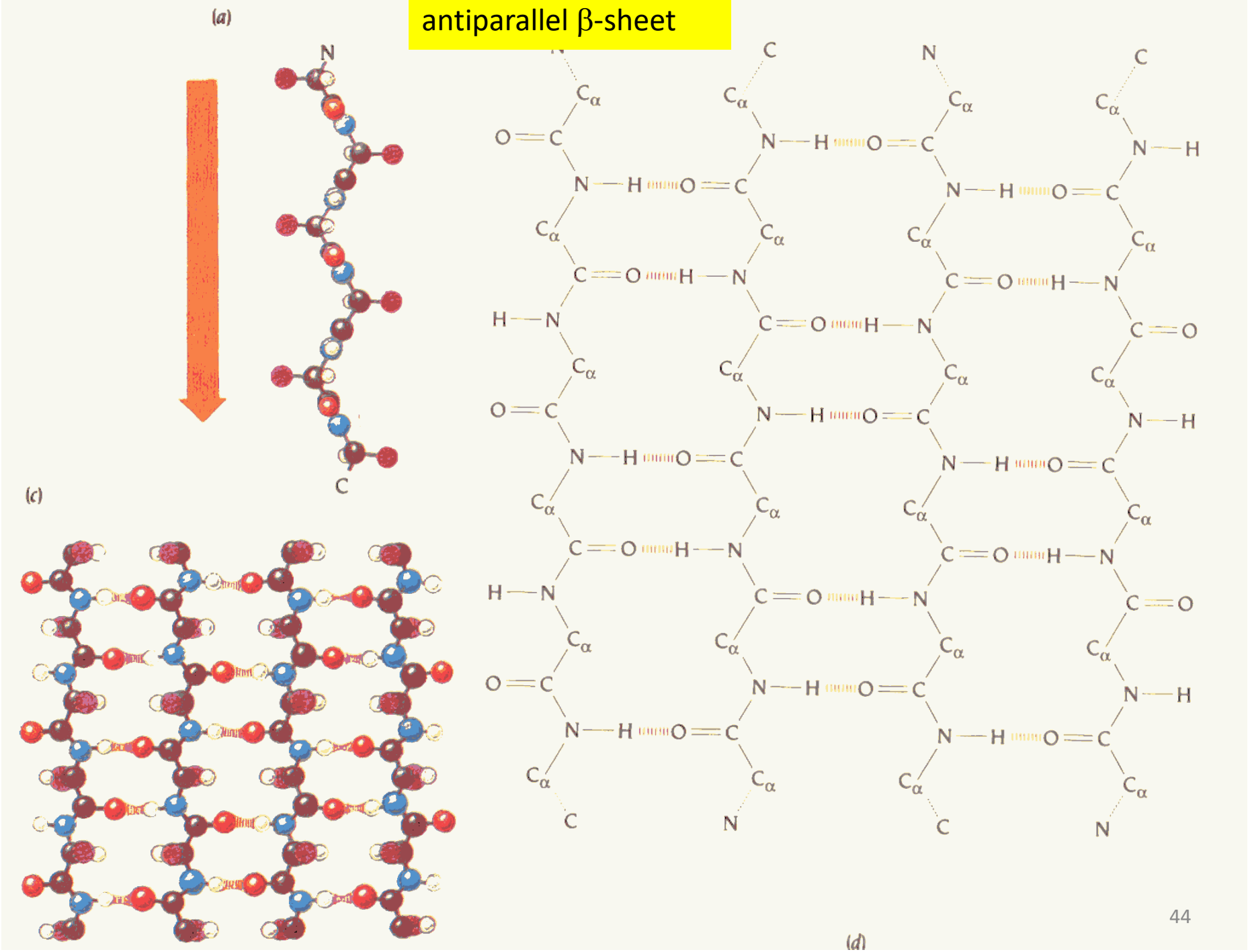
PRO, GLY, TYR, SER: poor α-helix formers

…weak tendency (not strong enough for prediction since hydrophobic residue can be on surface) for amino acids to change from hydrophillic to hydrophobic with a periodicity of 3-4 residues

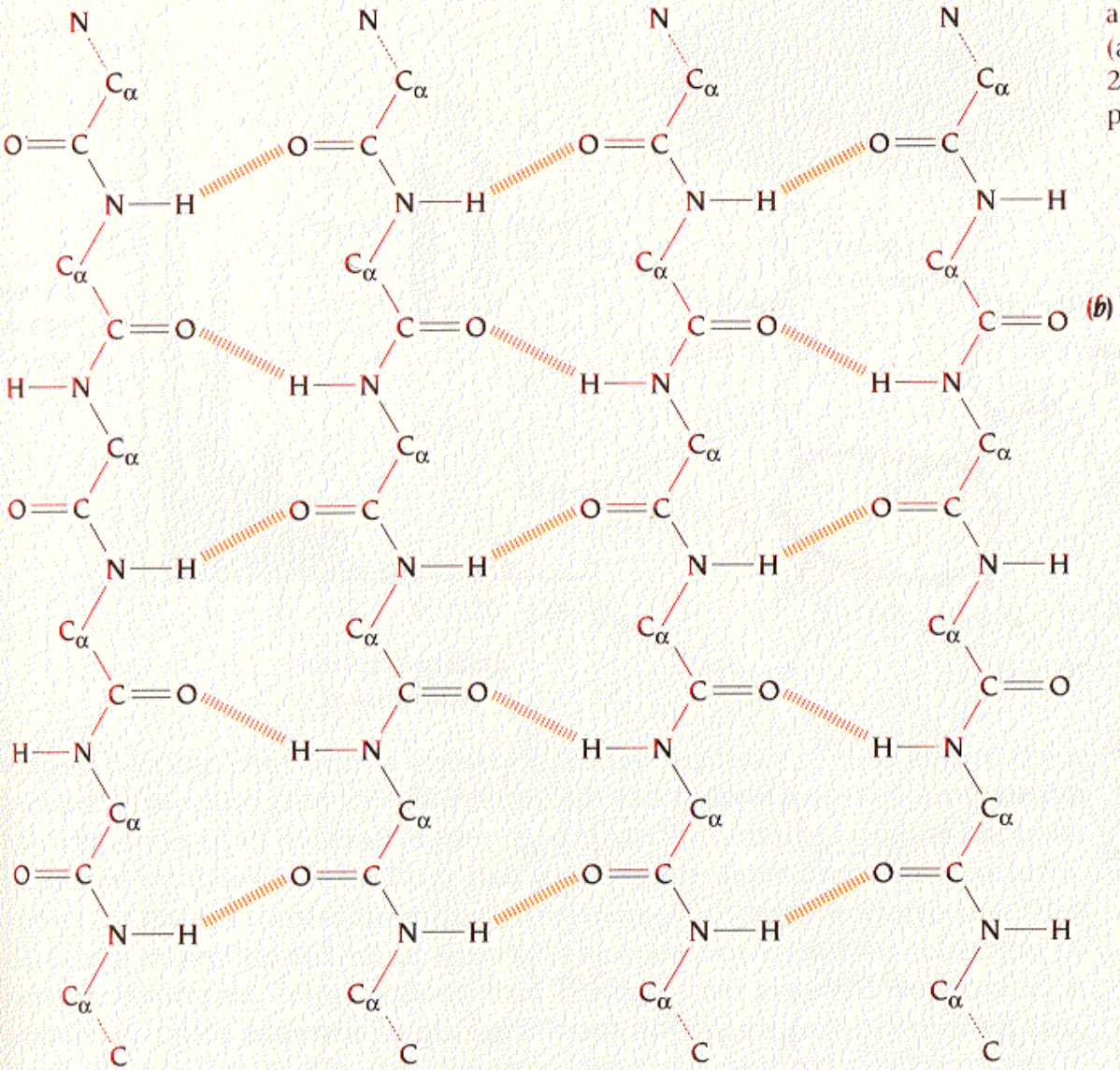
β-sheet structure

- built up from combinations of several regions of polypeptide chain
- β-strands are usually 5-10 residues long
- β-strands are usually almost fully extended
- β-strands are usually aligned adjacent to one another to form C=O to N-H bonds between the strands
- when several β-strands are together, this is called *pleated* (with C_α atoms above and below plane). antiparallel or parallel

antiparallel β-sheet



parallel β-sheet



(b)

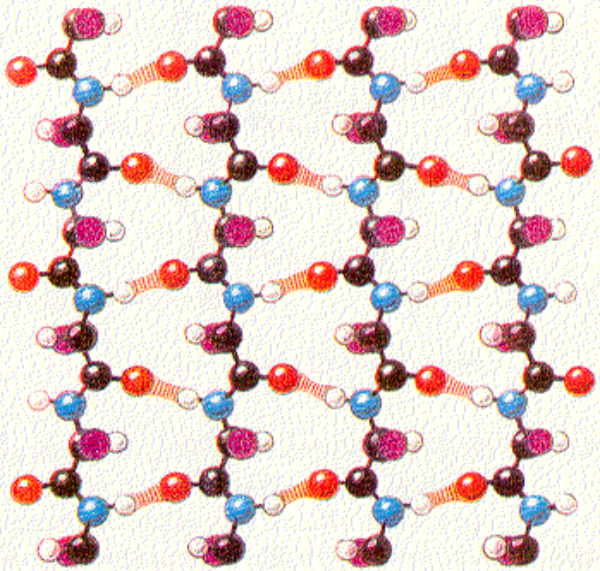
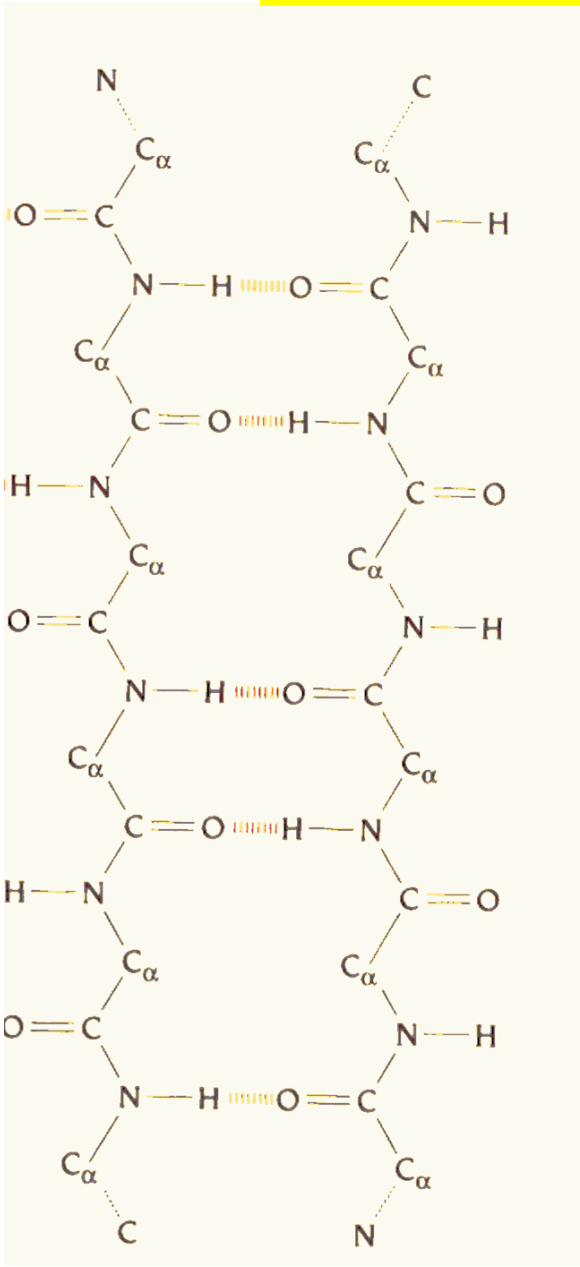
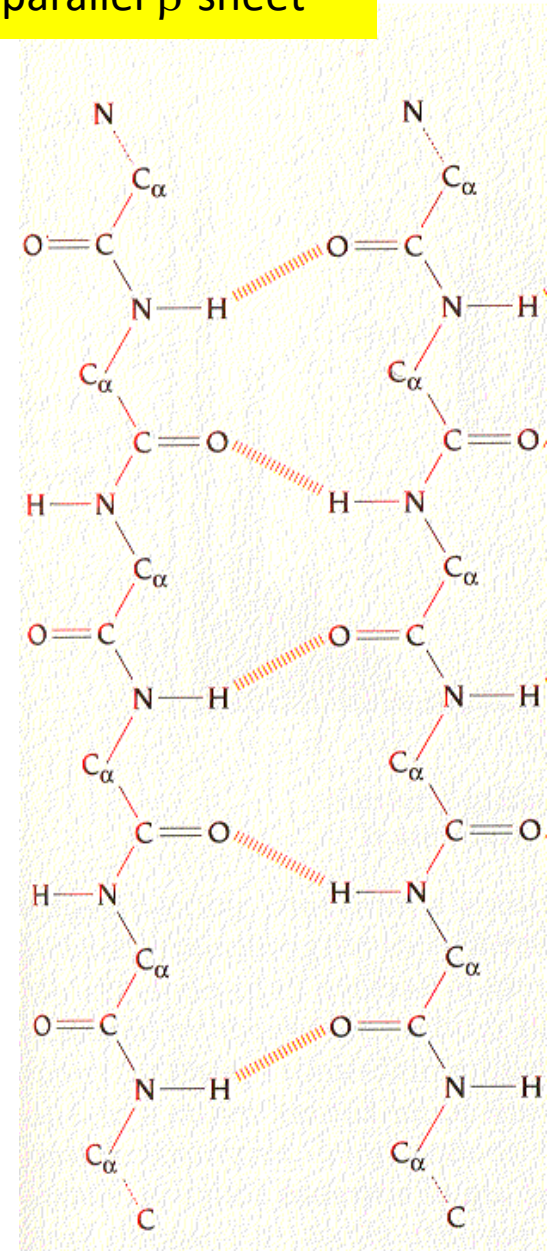


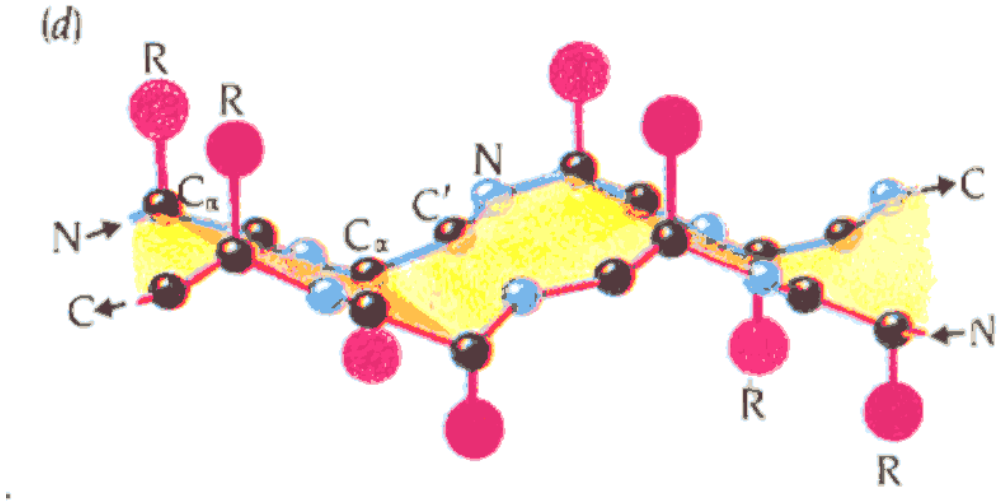
Figure 2.6 Parallel β sheet. (a) Schematic diagram showing the hydrogen bond pattern in a parallel β sheet. (b) Ball-and-stick version of (a). The same color scheme is used as in Figure 2.5c. (c) Schematic diagram illustrating the pleat of a parallel β sheet.

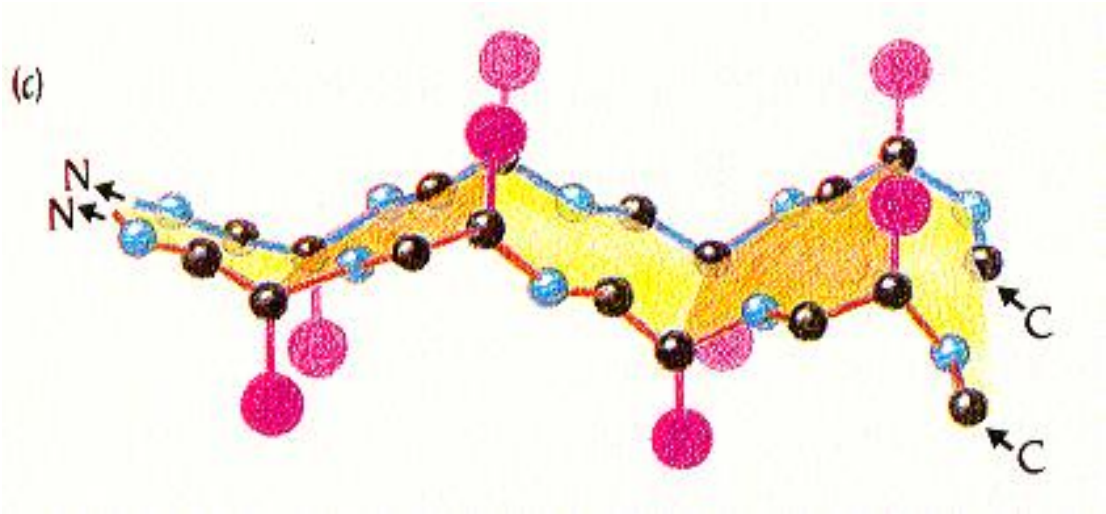
anti-parallel vs. parallel β-sheet





pleating in antiparallel vs. parallel β-sheet





loops and turns:

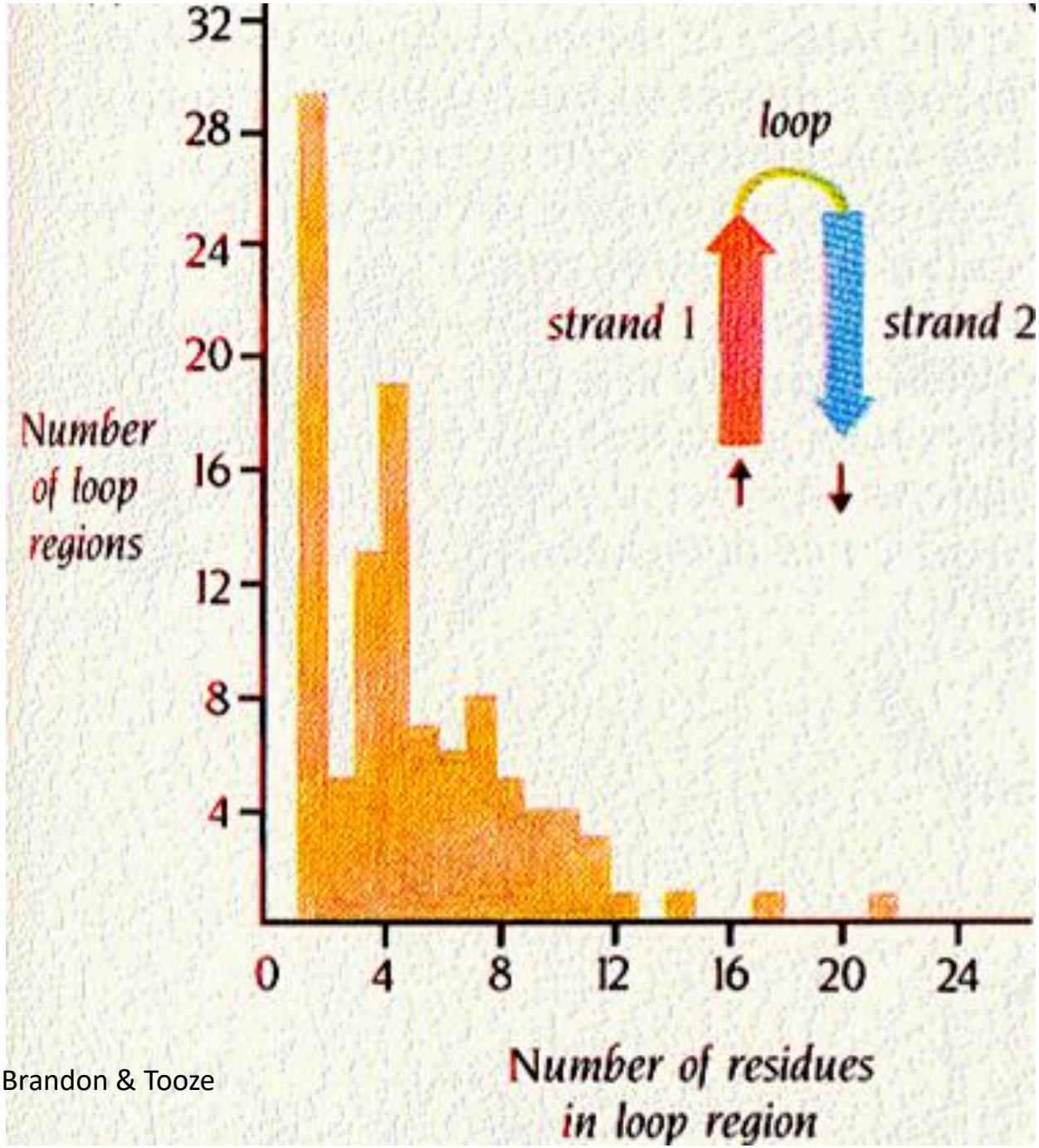


Figure 2.8, Brandon & Tooze

Loops and turns:

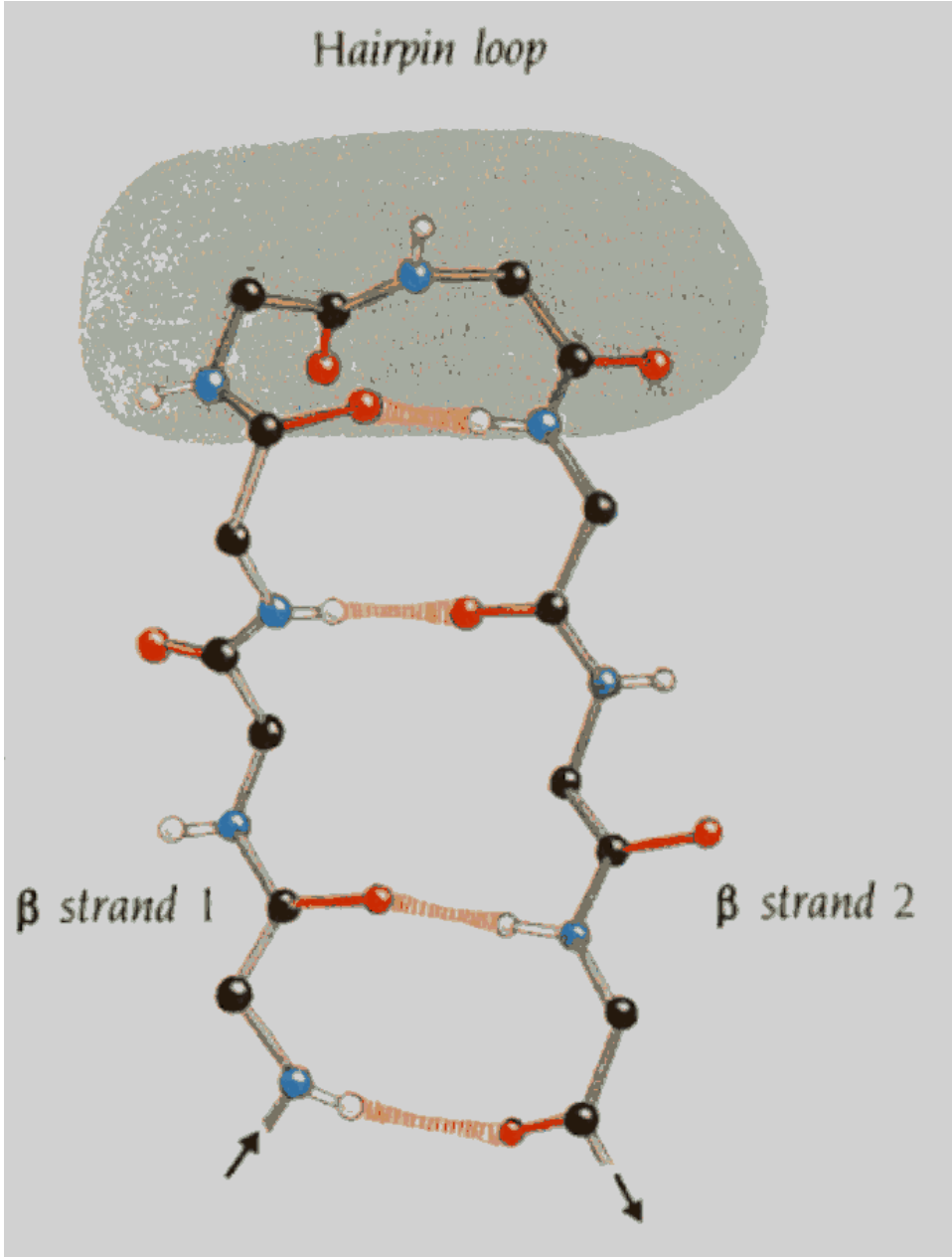
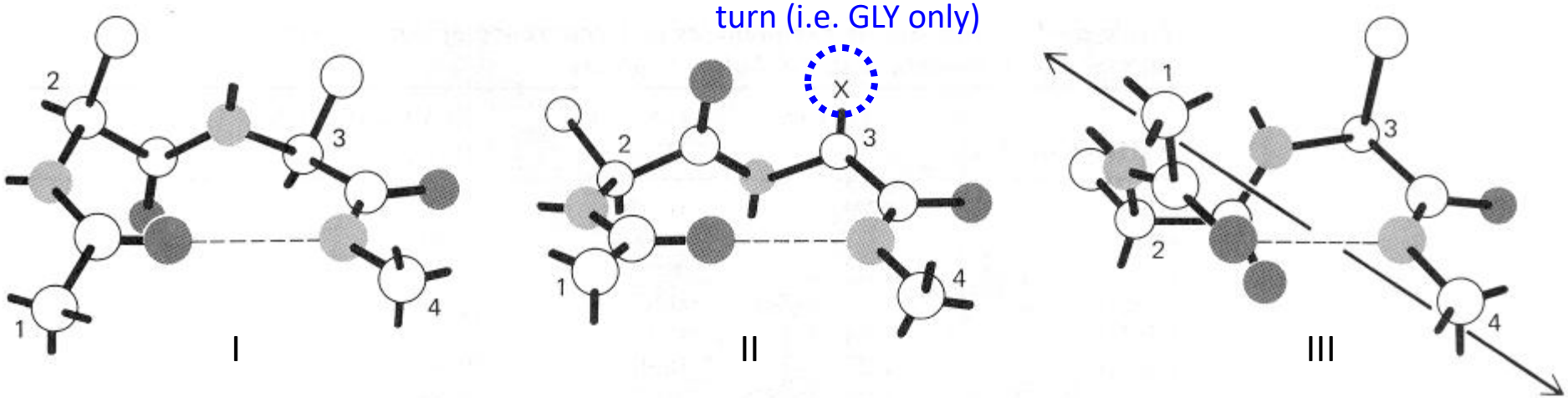


Figure 2.8, Brandon & Tooze

Classical types of reverse turns

Figure 6-18, Creighton, Proteins

no side chain may be present
on third residue in type II
turn (i.e. GLY only)



Resembles 3_{10} helix
(arrow denotes helical axis)

protein structural *motifs*

Simple combinations of secondary structural elements, called *motifs* or *supersecondary structure*.

Examples:

helix-turn-helix or helix-loop-helix

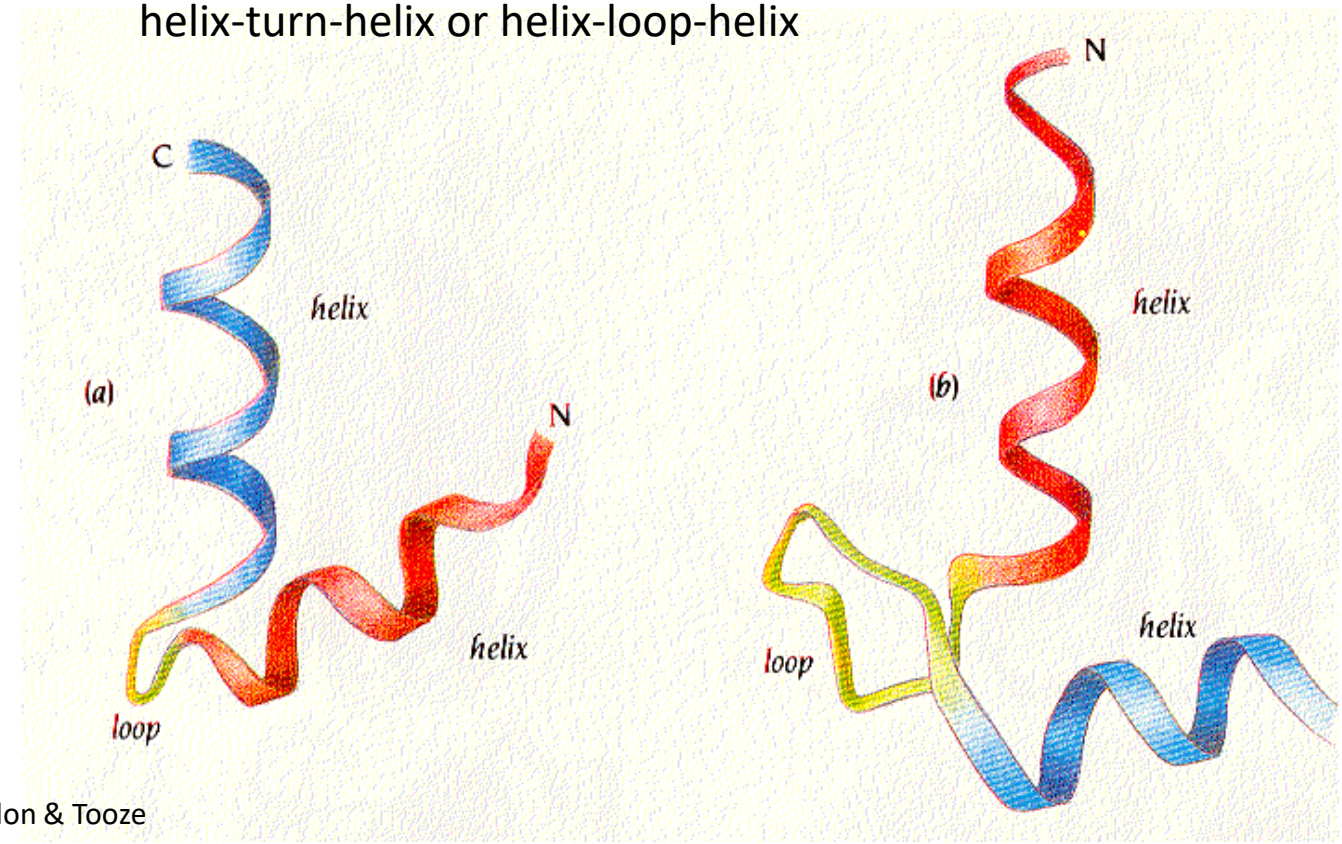


Figure 2.12 Brandon & Tooze

protein structural *motifs*

Simple combinations of secondary structural elements, called *motifs* or *supersecondary structure*.

Examples:

helix-turn-helix or helix-loop-helix

hairpin β

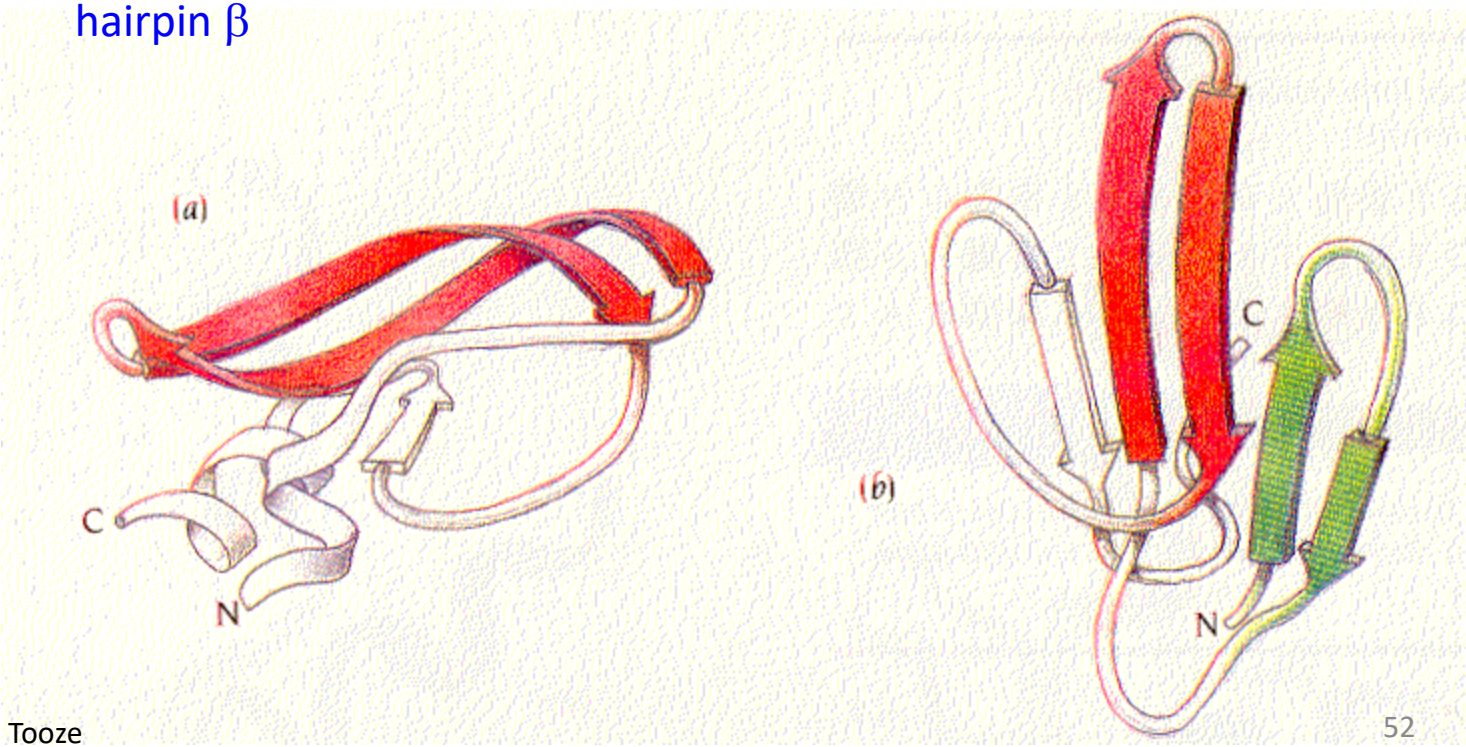


Figure 2.14 Brandon & Tooze

protein structural *motifs*

Simple combinations of secondary structural elements, called *motifs* or *supersecondary structure*.

Examples:

helix-turn-helix or helix-loop-helix

hairpin β

Greek key

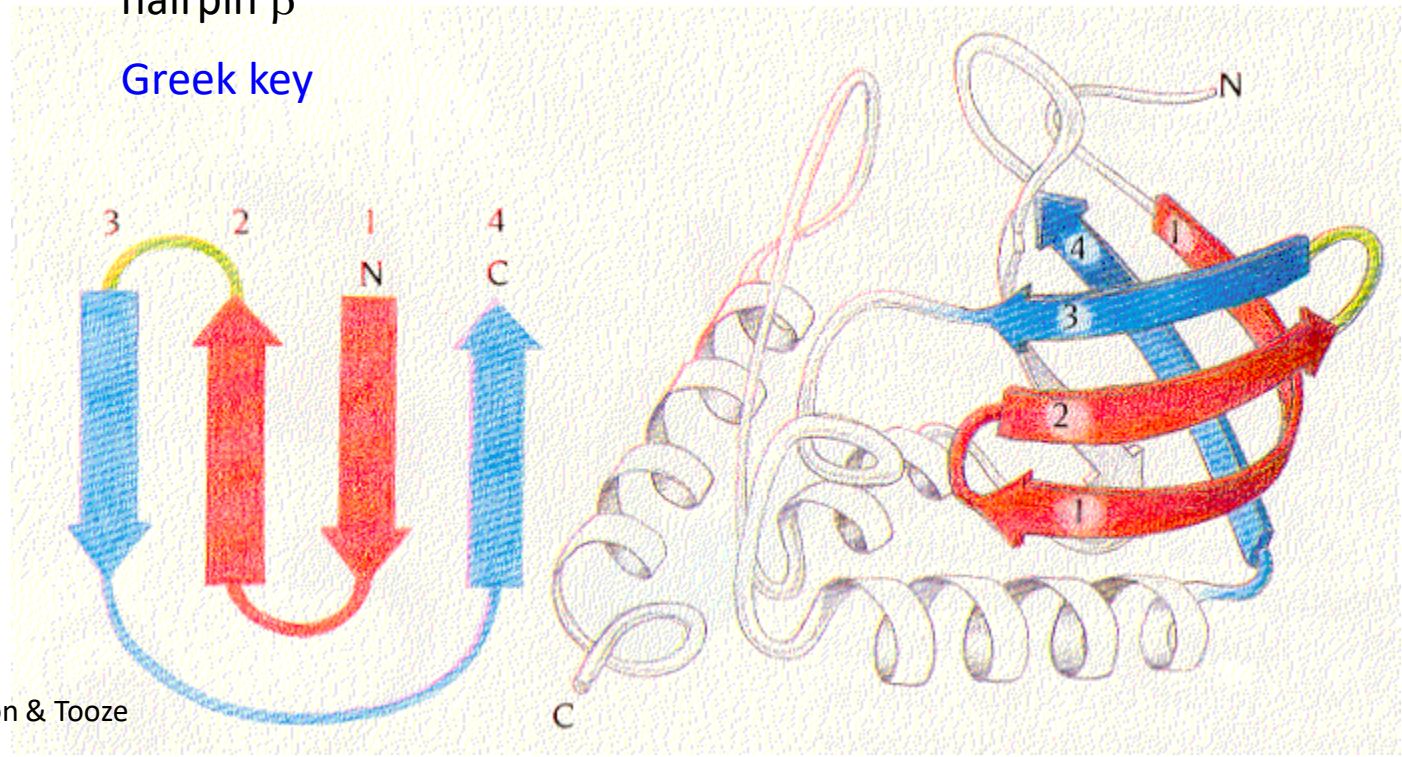


Figure 2.15 Brandon & Tooze

protein structural *motifs*

Simple combinations of secondary structural elements, called *motifs* or *supersecondary structure*.

Examples:

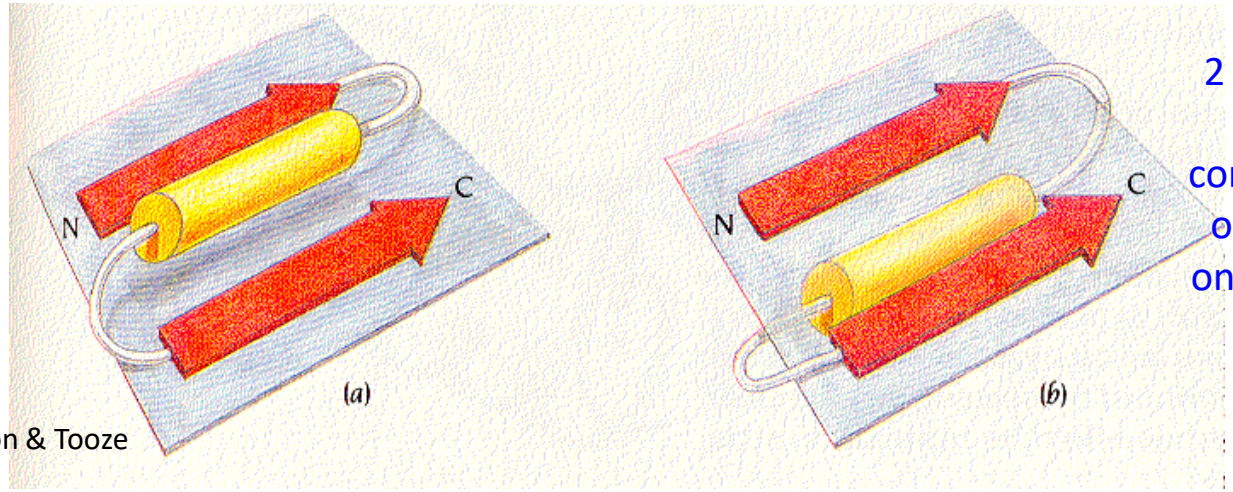
helix-turn-helix or helix-loop-helix

hairpin β

Greek key

beta-alpha-beta

Found almost in every protein structure with a parallel β-sheet



2 possible hands: left-handed connection (shown on the right) has only been found in 1-protein (subtilisin)

Figure 2.18 Brandon & Tooze

Domains are built from structural motifs

3 main classes of protein structures:

α-domains

β-domains (antiparallel β)

α/β domains

14729 PDB Entries or more. 35685 Domains. Literature References (excluding nucleic acids and theoretical models)

Class	Number of folds	# superfamilies	# families
All alpha proteins	144	231	363
All beta proteins	104	190	303
α/β proteins	107	180	409
$\alpha + \beta$ proteins	194	276	428
Multi-domain proteins	32	32	45
Membrane/cell surface	11	17	28
Small proteins	56	81	123
Total	648	1007	1699

Structural Classification Of Proteins

http://scop.mrc-lmb.cam.ac.uk/scop/

Root: scop

Classes:

1. All alpha proteins [46456] (218)
2. All beta proteins [48724] (144)
3. Alpha and beta proteins (a/b) [51349] (136)
   Mainly parallel beta sheets (beta-alpha-beta units)
4. Alpha and beta proteins (a+b) [53931] (279)
   Mainly antiparallel beta sheets (segregated alpha and beta regions)
5. Multi-domain proteins (alpha and beta) [56572] (46)
   Folds consisting of two or more domains belonging to different classes
6. Membrane and cell surface proteins and peptides [56835] (47)
   Does not include proteins in the immune system
7. Small proteins [56992] (75)
   Usually dominated by metal ligand, heme, and/or disulfide bridges
8. Coiled coil proteins [57942] (6)
   Not a true class
9. Low resolution protein structures [58117] (24)
   Not a true class
10. Peptides [58231] (116)
    Peptides and fragments. Not a true class
11. Designed proteins [58788] (42)
    Experimental structures of proteins with essentially non-natural sequenc

Hand-curated hierarchical taxonomy of proteins based on their structural and evolutionary relationships.

Classes

Fold Level

Super Families

Families

Superfamily

Family

CATH Protein Structural Data Base

http://www.biochem.ucl.ac.uk/bsm/cath/cath.html

CATH
Protein Structure Classification

CATH DHS Gene3D Impala FTP Internal

Search

Go!

- PDB Code
- CATH Code
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CATH Protein Structure Classification

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Options

- Browse or search the classification
- CATH statistics and release information
- General information on CATH
- CATH lists and FTP site
- DHS - Dictionary of Homologous Superfamilies. Summary of structural and functional features for CATH Homologous Superfamilies
- CATH File Formats (for FTP files)

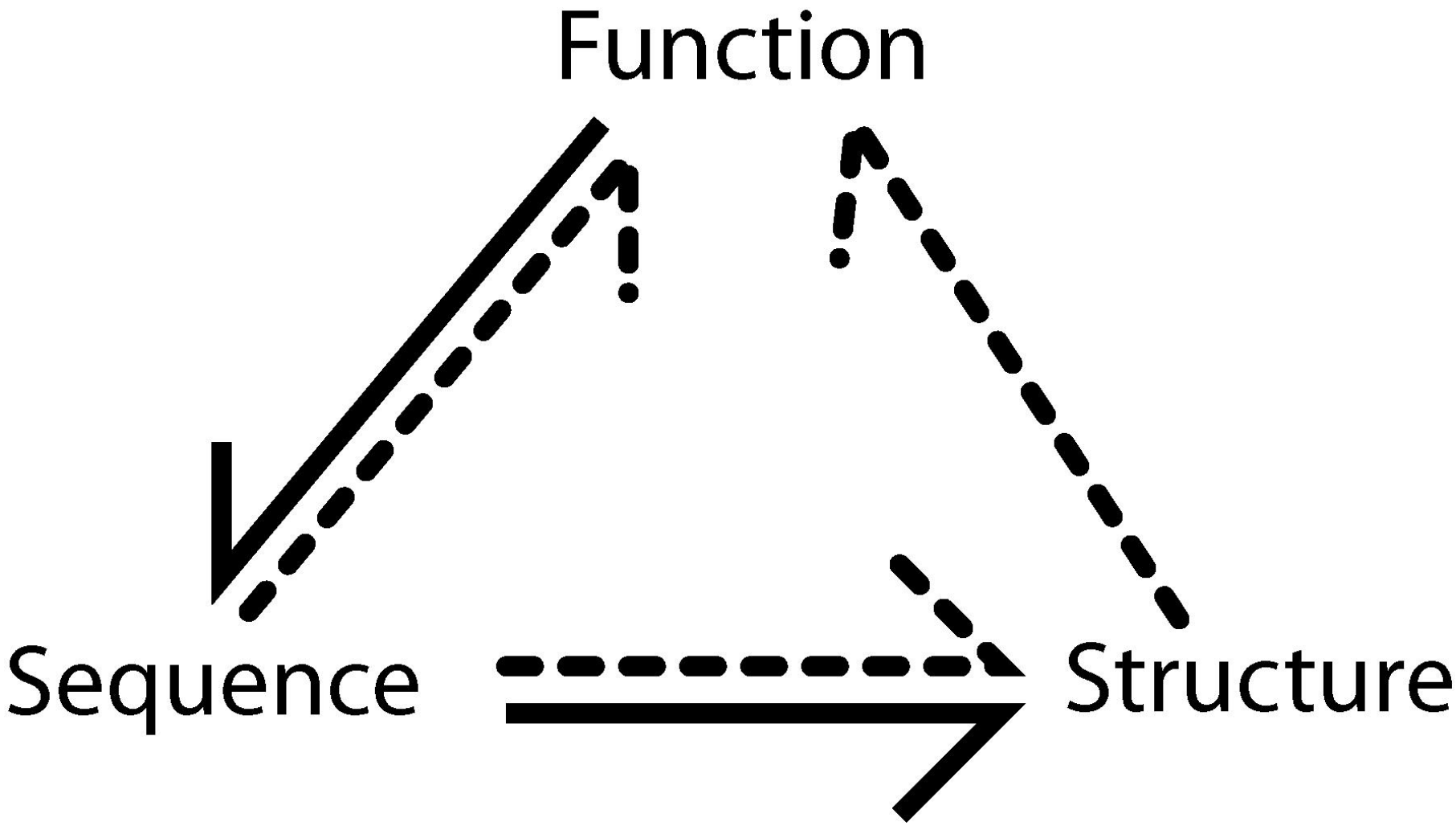
Introduction

CATH is a novel hierarchical classification of protein domain structures, which clusters proteins at four major levels, Class(C), Architecture(A), Topology(T) and Homologous superfamily (H).

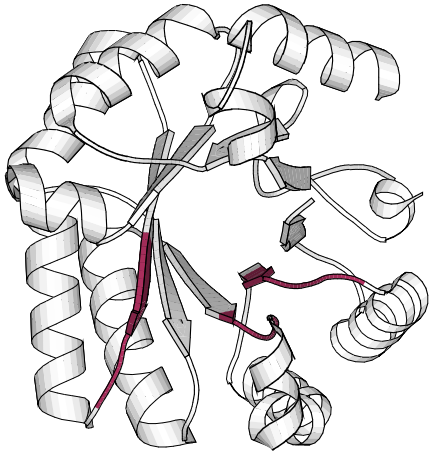
Semiautomatic domain classification of PDB crystal structures. Thornton (London)

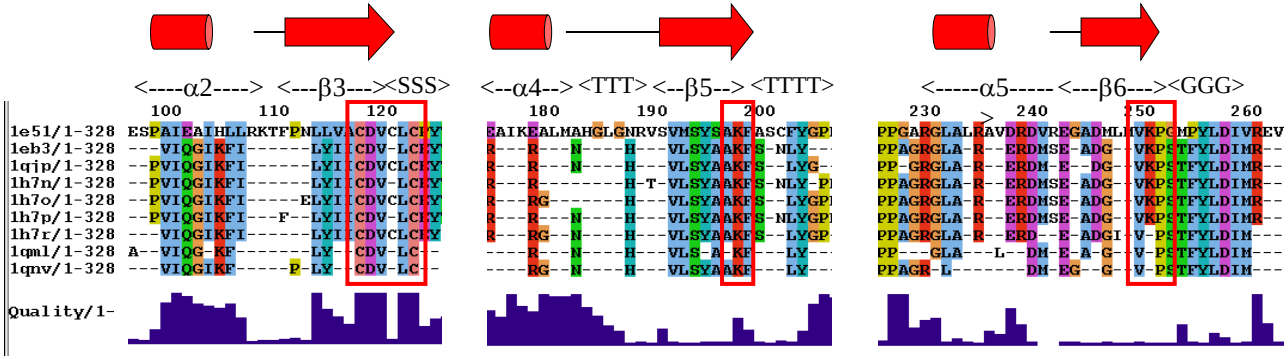
Class
Architecture
Topology
Homology
Sequence

Protein Information Resources


Function
Sequence
Structure

Relationship between Sequence, Function, & Structure

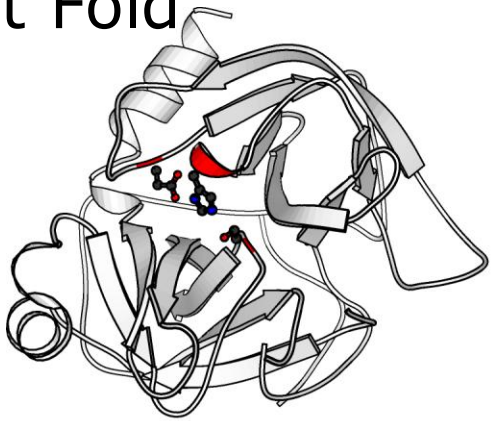




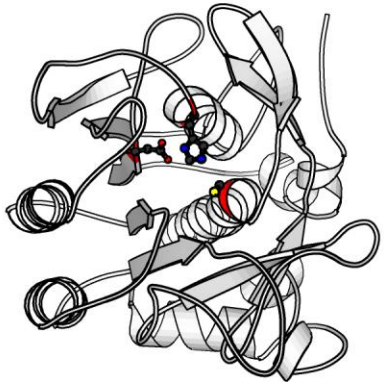
Problems with Structure Based Function Predictions

Similar Function
Different Fold

Chymotrypsin

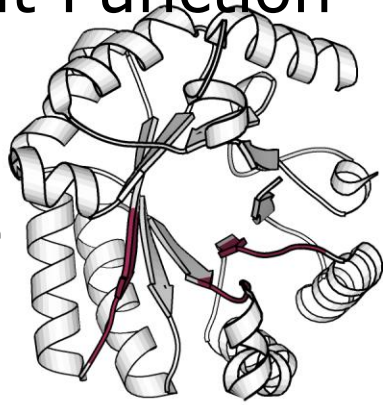


Subltilisin

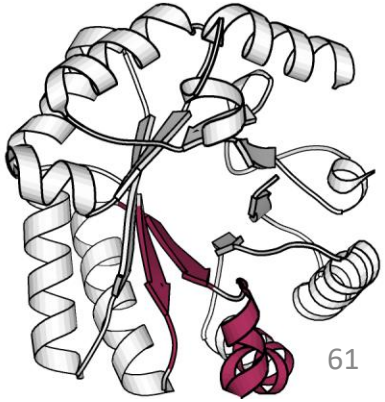


Similar Fold
Different Function

Dehydratase



Hydrolase



Proteomics

- **What** is PROTEOMICS?
  - -OMICS has become the suffix to denote the study of the entire set of something
  - Genomics: study of all genes
  - Proteomics: study of all proteins
  - Transcriptomics: study of all mRNA transcripts
  - Metabolomics: study of metabolites in cell

Proteomics

Proteomics questions

- Which proteins are made from the genome?
- What is their 3D structure?
- Where they are?
- What they do?
- Which other proteins they interact with?
- Are they modified in the cell post-translationally?

Proteomics

Key proteomic technologies

- 3D structure determination (X-ray/NMR)
- 2D Gels to assess all the proteins in a cell.
- Mass spectrometry to identify proteins, protein modifications.
- Yeast-Two-Hybrid systems to assess protein-protein interactions
- Protein Arrays to assess all proteins in a cell using antibodies or other recognition technology.

Proteomics

Proteome: a genome's complete complement of expressed proteins

Protein related data

Sequence - SWISS-PROT database http://www.expasy.ch/sprot/sprot-top.html

Structure - Protein Data Bank http://www.rcsb.org

Expression - DNA & Protein expression arrays

Function - Structural Genomics

Proteomics Tools (Sequence)

Secondary Structure Prediction

- PredictProtein - Neural Net
  http://maple.bioc.columbia.edu/predictprotein/
- NNPredict - Neural Net
  http://www.cmpharm.ucsf.edu/~nomi/nnpredict.html
- Jpred - Consensus of Predictions
  http://jura.ebi.ac.uk:8888/
- SAMT99 - Hidden Markov Model
  http://www.cse.ucsc.edu/research/compbio/HMM-apps/HMM-applications.html

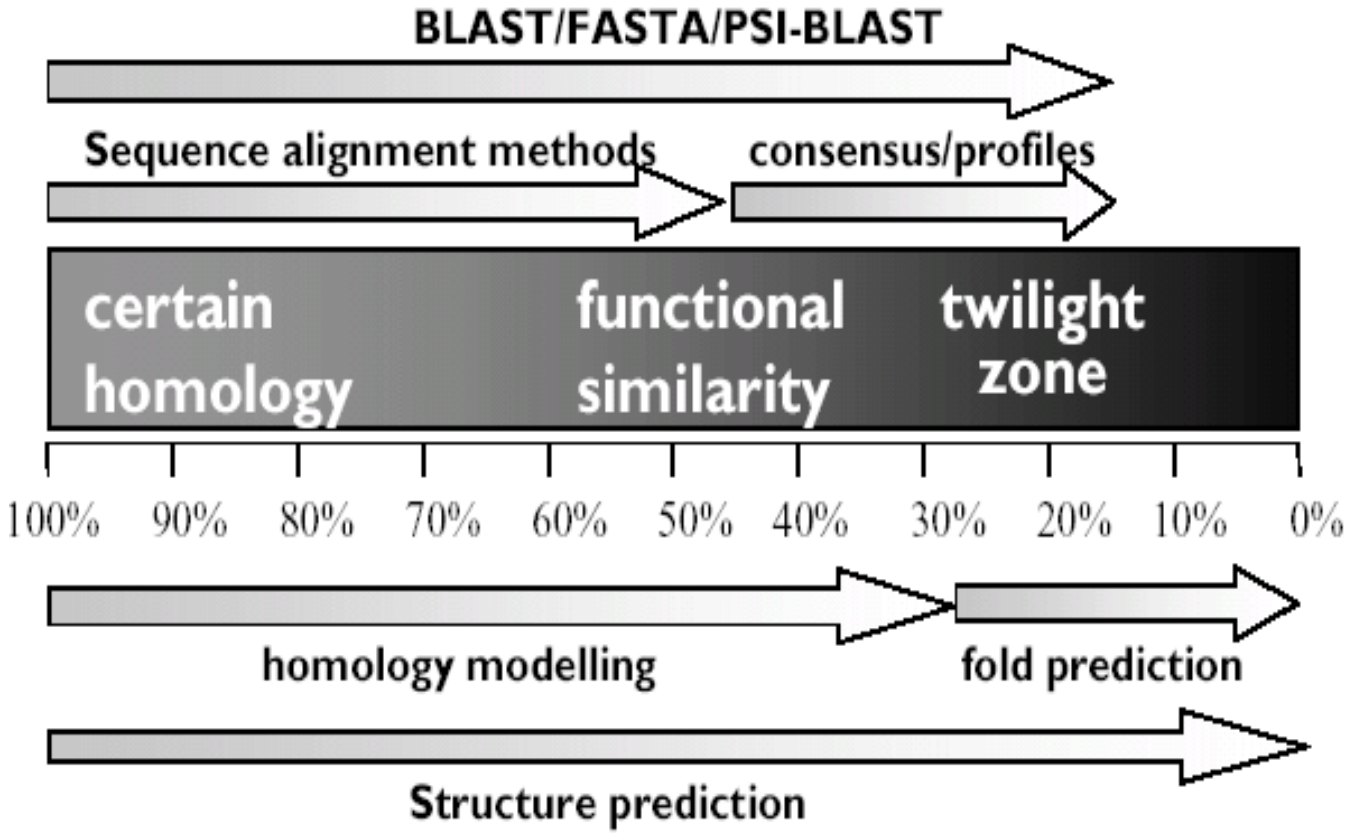
Anatomy of a Protein Structure File

- Header: Author, Experiment, Sequence, other information
- Atoms: Protein Coordinates
- Ligands: HetAtms
- Problems

Protein Research

- Protein Classification and Structure Prediction e.g. for drug design
  - The relationship between sequence, structure and function of proteins is not always straightforward
    - Many challenges remain open in the study of proteins functions, folding, 3D structure, ….
  - Databases are composed which collect essential features from the sequence families in the form of domain or motif descriptors
    e.g. PROSITE
  - The use of motifs has proven to be very useful for the functional assignment of unknown proteins

Sequence Similarity and Tools



Sequence-function relationship

- Swiss-prot is a database containing most of the known proteins with annotated biological functions

- For a unknown gene, we could possibly derive its function by finding proteins in Swiss-prot that are homologous to the gene

- For a newly sequenced microbial genome, at least 50% of the genes can have some level of functional predictions through sequence comparison against databases like Swiss-prot

Sequence-function relationship

- Sequence alignment between two sequences can help to determine if genes A and B are homologous and hence gene A has similar biological function of gene B

- Sequence alignment between multiple sequences can help to find functional sites (e.g., active sites) in a protein sequence

```
SAANLEYLKNVLLQFIFLKPG--SERERLLPVINTMLQLSPEEKGKLAAV
NEKNMEYLKNVFVQFLKPESVP-AERDQLVIVLQRVLHLSPKEVEILKAA
KNEKIAYIKNVLLGFLEHKE----QRNQLLPVISMLLQLDSTDEKRLVMS
REINFEYLKHVVLKFMSCRES---EAFHLIKAVSVLLNFSQEEENMLKET
EPTEFEYLRKVMFEYMMGR-----ETKTMAKVITTVLKFPDDQAQKILER
DPAEAEYLRNVLYRYMTNRESLGKESVTLARVIGTVARFDESQMKNVISS
STSEIDYLRNIFTQFLHSMGSPNAASKAILKAMGSVLKVPMAEMKIIDKK
```

Conserved motifs in multiple sequence alignments tend to be functional sites as there is a pressure to keep them un-mutated during evolution while others part have less pressure to do so

Sequence-structure relationship

- Sequence similarity often implies structural similarity (though exceptions exist)

- If protein A and protein B have similar sequences and protein A has its structure solved, protein B's structure could be (very) similar to protein A's (and hence make B's structure prediction)

```
SAANLEYLKNVLLQFIFLKPG-SERERLLPVINTMLQLSPEEKGKLAAV
NEKNMEYLKNVFVQFLKPESVPAERDQLVIVLQRVLHLSPKEVEILKAA
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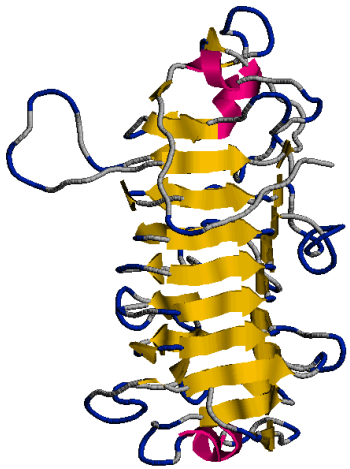
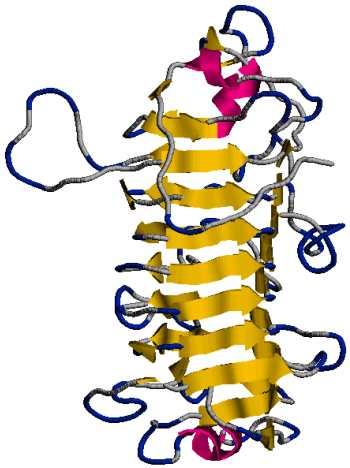


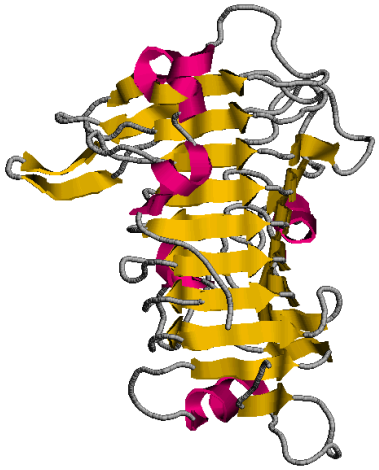
Protein A

proteins A and B have high sequence similarity

prediction



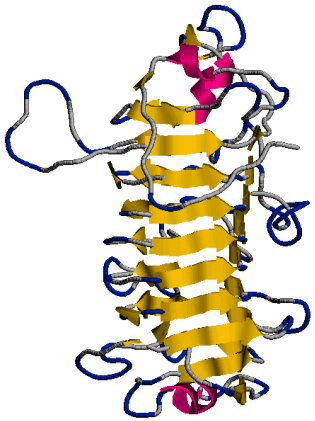
Protein B

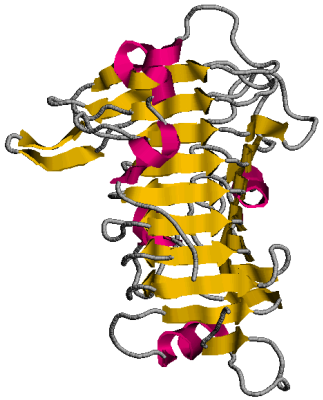


Actual Structure of B

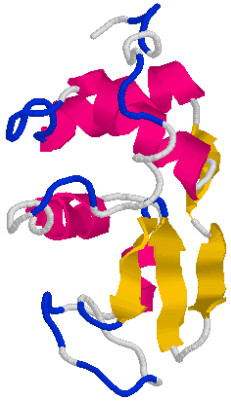
Sequence-structure relationship

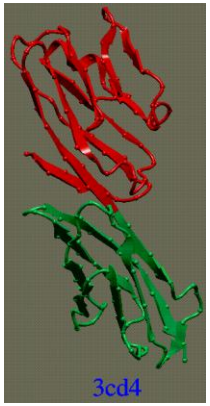
- Structural similarity

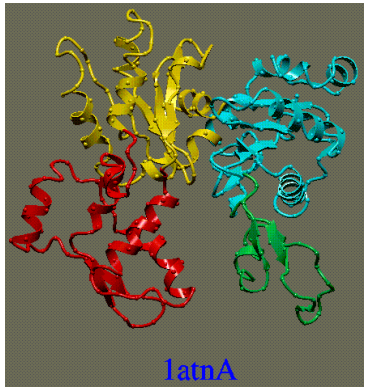


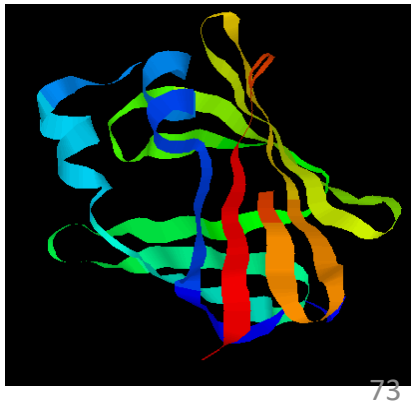


similar structures









Sequence-structure relationship

- Why do we want to know protein structures?
- Structures provide how a protein carries out its biological function
- Knowing the shape of the binding site can help design inhibitors – applications to drug design
- Knowing the positions of functional sites can help design experiments for investigation of functional mechanisms of a protein

Sequence comparison

- Sequence comparison is highly important tool for functional and structural studies of proteins

Generating Nice Figures

Protein Structure Visualization Programs

!! Free !!

- Rasmol
- Pymol
- Jmol
- SwissPDBview & Pov-Ray

http://www.expasy.ch/spdbv/text/pov.htm